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Dynamical System Analysis and Modeling of Early-Stage Thermal Runaway in Lithium-Ion Batteries

Xu Yuyang



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Abstract

Thermal runaway induced by internal short circuit in lithium-ion batteries is not only a fixed temperature threshold problem, but a nonlinear dynamical process affected by short circuit heat generation, heat dissipation, chemical side reactions, and state of charge depletion. This thesis develops an early stage thermal safety assessment framework based on a two dimensional temperature-SOC ODE model. The model includes internal short circuit ohmic heat, external current heat, early solid electrolyte interphase and anode reaction heat, and equivalent linear heat dissipation. Based on the model, a parameter dependent time scale ratio $\epsilon(R_{isc}, U_{eq})$ is derived to describe the relative speed between thermal response and SOC depletion, and to divide the (R_{isc}, U_{eq}) parameter plane into different dynamical regions. In the strongly separated region, the critical manifold and fast thermal attraction are analyzed. In the weak and non fast slow regions, a Jacobian based local thermal stability criterion is used. The numerical simulations are then used to compare the 70°C warning boundary and the 120°C stopping boundary with the ϵ curves. Sensitivity analyses show that ambient temperature strongly shifts the thermal boundaries, while external current mainly affects SOC depletion and the time scale separation. The proposed framework connects dynamical systems analysis with early stage battery safety judgment and provides an interpretable criterion for BMS applications.

Keywords: lithium-ion battery; thermal runaway; dynamical systems; time scale separation; battery management system

Faculty of Science and Technology

Uppsala University

Supervisor: Denis Gaidashev Subject reader: Yicun Huang

Examiner: Valeria Castellucci

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1 Introduction

This chapter, as the opening part of the thesis, mainly introduces the research background, motivation, and scope of this study on thermal runaway induced by internal short circuit in lithium ion batteries. Thermal runaway induced by internal short circuit is not simply a temperature threshold problem. Instead, it is a nonlinear dynamical process jointly determined by short circuit heat generation, heat release from chemical side reactions, heat dissipation conditions, and the evolution of state of charge (SOC). In particular, during the early stage of internal short circuit, the system may exhibit both fast temperature response and slow SOC depletion, while under severe short circuit conditions, it may also lose clear time scale separation. Therefore, a single temperature threshold is not sufficient to cover all internal short circuit conditions.

Specifically, Section 1.1 introduces the research background of lithium ion battery thermal runaway and explains the special role of internal short circuit in the early evolution of thermal runaway. Section 1.2 reviews current methods for thermal runaway warning, internal short circuit diagnosis, and battery thermal modeling, including empirical threshold methods, model driven methods, data driven methods, and multi sensor fusion methods. It also analyzes the limitations of these methods in explaining dynamical boundaries under different internal short circuit intensities and heat dissipation conditions. Section 1.3 presents the main research idea of this thesis: establishing a two dimensional temperature–state of charge (T–SOC) thermal runaway model, deriving a time scale parameter related to internal short circuit resistance and heat dissipation capability, and using fast subsystem analysis, local thermal stability analysis, and simulation based thermal boundary comparison in different time scale regions. Finally, Section 1.4 gives the organization of the thesis.

1.1 Background and Motivation

In recent years, the rapid development of electric vehicles and energy storage systems has greatly promoted the application of high energy density lithium ion batteries. However, battery safety issues, especially thermal runaway dominated by strongly exothermic chain reactions, remain one of the key problems limiting the further development of this industry^[1]. Thermal runaway is usually accompanied by a sharp temperature rise, the release of flammable and toxic gases, and, in severe cases, even combustion and explosion, posing serious threats to passenger safety and property safety.

Among various triggers of thermal runaway, internal short circuit is a fault form that is highly hidden, difficult to identify at an early stage, and associated with serious consequences^[2]. Under actual operating conditions, manufacturing defects, lithium dendrite growth, local separator damage, or slight mechanical deformation may all induce early stage internal short circuit. Unlike external abuse conditions, early internal short circuit often only appears as a weak short circuit current and slow local heating, while the terminal voltage and surface temperature may still remain within a normal range. Therefore, before an internal short circuit truly develops into a severe fault, it is often not easy to capture directly through traditional monitoring signals.

From the perspective of battery management system (BMS), the most valuable task is not the identification after thermal runaway has fully occurred, but the early judgment before thermal runaway enters an irreversible chain reaction. In the early stage of internal short circuit, for example, when the battery temperature has not yet reached the conventional high temperature alarm level, chemical side reactions are usually not yet the dominant heat source. The system behavior is mainly determined by the competition among internal short circuit ohmic heat, early chemical heat, and heat dissipation capability^[3]. If the BMS can determine whether the current operating condition has sufficient thermal stability margin at this stage, it may have the opportunity to block the fault evolution by disconnecting the circuit or enhancing cooling. Conversely, if the system is already approaching some critical boundary, stronger protection should be triggered in advance, rather than waiting until the temperature reaches a fixed alarm threshold.

An important feature of internal short circuit induced thermal runaway is that its time scale is not fixed. Under soft short conditions, SOC depletion is relatively slow, while temperature may approach a certain thermal equilibrium state relatively quickly. However, under severe short circuit conditions, the short circuit current increases significantly, the SOC depletion rate becomes much faster, and the temperature evolution and SOC evolution may be on the same order of magnitude. In other words, whether the system has a fast slow structure itself depends on parameters such as internal short circuit resistance, heat dissipation capability, and battery capacity, and cannot simply be preset as a single time scale system or a fixed fast slow system.

This is exactly the motivation of this thesis. This work does not treat early warning of thermal runaway simply as the question of whether the temperature exceeds a fixed threshold. Instead, it views the T–SOC evolution under internal short circuit conditions as a parameter dependent nonlinear dynamical system. By establishing a two dimensional ordinary differential equation (ODE) model for early stage internal short circuit, this thesis

aims to provide a more interpretable early stage thermal safety criterion from the perspectives of phase space structure, time scale partitioning, and stability boundaries. The goal of this method is not to replace all existing BMS diagnostic methods, but to provide a criterion that can quickly determine the thermal risk level for a BMS when the internal short circuit resistance R_{isc} and heat dissipation capability U_{eq} have been estimated.

In an actual BMS architecture, the identification of early internal short circuit can usually be completed by upstream state estimation algorithms. However, merely identifying the existence of internal short circuit does not mean that the safety decision has been completed. The BMS still needs to further determine whether, under the current R_{isc} , SOC, and heat dissipation conditions, this internal short circuit will drive the system into a high temperature warning region, cross the safety boundary of the model, or remain in a controllable thermally stable region. This is the core question that the subsequent dynamical systems analysis in this thesis aims to answer.

1.2 State of the Art and Limitations

Around the prediction and early warning of lithium ion battery thermal runaway, a large number of studies have been carried out from the perspectives of experimental features, mechanism models, data driven methods, and multi sensor monitoring^[4]. These methods have their own advantages in different scenarios, but for the dynamical stability judgment in the early stage of internal short circuit, there are still some issues worth further investigation.

The most direct type of method is threshold warning based on signals such as temperature, voltage, and current. This type of method has a simple structure, strong real time performance, and is easy to deploy in engineering BMSs. Therefore, it is still an important part of practical safety protection strategies. However, fixed thresholds often fail to reflect the dynamical differences under different internal short circuit intensities and different heat dissipation capabilities. For example, the same temperature value may correspond to completely different subsequent risks under soft short and severe short conditions. Similarly, the same R_{isc} may also lead to different temperature evolution results under strong and weak heat dissipation environments. Therefore, although a simple threshold method is suitable for rapid protection, it is difficult to provide a structural explanation of the thermal instability boundary.

Mechanism driven modeling methods can more directly describe the thermal, electrical, and chemical reaction processes inside the battery. High fidelity three dimensional multi physics models can describe the internal temperature distribution, electrochemical reactions, and spatial evolution of local short circuit hot spots through partial differential equations^[5]. Such models have high physical accuracy and can capture the effects of local hot spots, spatial temperature differences, and internal short circuit locations on thermal runaway. However, they involve many parameters and high computational cost, making them difficult to use directly for fast BMS judgment. In comparison, lumped ODE based thermal models have lower computational cost. They often use accelerating rate calorimetry (ARC) and other experiments to obtain chemical reaction activation energies, pre exponential factors, and reaction heats^[6], and are suitable for parameter scanning and safety boundary analysis. However, many thermal runaway kinetic models focus more on the temperature rise after thermal runaway has been triggered, while the BMS is more concerned with the early warning window before thermal runaway. For early internal short circuit conditions, merely predicting the temperature trajectory is not sufficient. It is also necessary to further determine whether the system will lose stable thermal equilibrium, whether it will enter the warning temperature region, and whether different mathematical analysis are applicable under different time scales.

Thermal runaway warning methods based on data driven and multi sensor warning can extract nonlinear features from complex operating conditions^[7]. These methods provide new ideas for engineering early warning, but they also face problems such as scarce fault samples, difficulty in obtaining extreme thermal runaway data, complex sensor arrangement, and insufficient model interpret ability. Especially in the early stage of internal short circuit, real fault samples are even scarcer. Therefore, models with clear physical meaning are still needed to complement data driven methods.

For internal short circuit itself, existing studies have systematically summarized its triggering mechanisms, modeling methods, detection methods, and prevention strategies. Existing internal short circuit diagnosis methods can usually estimate or identify the existence and severity of internal short circuit. For example, internal short circuit resistance R_{isc} can be obtained through equivalent circuit models, electro thermal models, state observers, or data driven anomaly detection^[8]. However, after R_{isc} is obtained, the BMS still needs to answer a further question: is the current internal short circuit sufficient to drive the system across the thermal safety boundary? This question is not simply a fault identification problem, but a dynamical systems stability problem jointly related to internal short circuit intensity, SOC, heat dissipation capability, and chemical heat

release.

To summarize, the existing literature provides important foundations for understanding thermal runaway mechanisms, internal short circuit diagnosis, battery thermal modeling, and early warning methods. However, most threshold based methods do not explain how the thermal risk changes with internal short circuit resistance and heat dissipation strength. High fidelity multiphysics models can describe the physical process in more detail, but they usually require many parameters and high computational cost, which makes them difficult to use directly for fast parameter plane safety judgment. Data driven warning methods can extract useful features from operating data, but their physical interpretation is still limited when internal short circuit fault samples are scarce.

Therefore, the research gap addressed in this thesis is not the detection of internal short circuit itself, nor the full simulation of thermal runaway propagation after the cell has entered the severe high temperature stage. Instead, the focus is on the following question: after an internal short circuit resistance R_{isc} and an equivalent heat dissipation coefficient U_{eq} have been estimated, can these two quantities be used to construct an interpretable dynamical safety judgment in the (R_{isc}, U_{eq}) parameter plane? This thesis addresses this gap by connecting a reduced two dimensional T-SOC model, a parameter dependent time scale ratio, local thermal stability analysis, and simulated thermal safety boundaries.

1.3 Research Objectives and Main Contributions

The overall objective of this thesis is to establish an interpretable early stage thermal safety assessment method for internal short circuit (ISC) in lithium ion batteries, with potential application in battery management systems (BMS). Different from a fixed temperature threshold method, this thesis aims to explain how the thermal response depends on the internal short circuit resistance, heat dissipation capability, external current, ambient temperature, and the coupled evolution of temperature and state of charge (SOC). The main question is not only whether the cell temperature reaches a prescribed threshold, but also how the system location in the (R_{isc}, U_{eq}) parameter plane is related to time scale separation and thermal safety boundaries.

The main contributions of this thesis can be summarized as follows.

First, a two dimensional T-SOC dynamical model is established for early stage ISC analysis. The model includes internal short circuit ohmic heat, external current heat, solid electrolyte interphase (SEI) and anode reaction heat, and equivalent heat dissipation. The internal short circuit resistance R_{isc} and the equivalent cooling coefficient U_{eq} are treated as key parameters that determine the thermal response and the structure of the parameter space.

Second, a parameter dependent time scale ratio

$$\epsilon(R_{isc}, U_{eq}) \quad (1)$$

is derived to describe the relative relationship between the thermal response time and the SOC depletion time. Based on this parameter, ISC conditions are classified into different time scale regimes, including a strong fast slow separation region, a weak fast slow separation region, and a non fast slow separation region. This avoids applying the same mathematical criterion to all parameter regions.

Third, the relationship between simulated thermal boundaries and time scale contours is analyzed in the (R_{isc}, U_{eq}) parameter plane. The simulated 70°C warning boundary and the 120°C stopping boundary are compared with corresponding constant- ϵ curves. The mean time scale ratio and its relative variation along each thermal boundary are used to quantify how closely a thermal boundary can be represented by a single characteristic time scale value.

Fourth, sensitivity analyses with respect to ambient temperature and external current are performed to examine the robustness of the proposed boundary interpretation. These simulations show how changes in operating conditions shift the thermal response and modify the location of warning and high temperature regions. This provides a more practical interpretation of the proposed method for early stage ISC warning under different thermal and electrical operating conditions.

Overall, the thesis provides an interpretable framework that connects dynamical system analysis, time scale separation, and simulated thermal safety boundaries. The proposed method does not wait until the cell temperature has already reached a high risk threshold. Instead, once R_{isc} , U_{eq} , ambient temperature, and external current conditions are estimated, the corresponding location in the parameter plane can be used to infer the possible later thermal tendency of the system.

It should be clarified that this thesis does not propose new electrochemical reaction kinetics or new experimental battery parameters. The OCV–SOC relation, the early chemical reaction structure, the kinetic parameters, the internal short circuit heat efficiency factor, and the lumped thermal modeling assumptions are adopted from literature based models. The contribution of this thesis is to combine these elements into a two dimensional T–SOC dynamical system for early stage ISC analysis, to introduce a parameter dependent time scale ratio $\epsilon(R_{isc}, U_{eq})$, and to use time scale partition, local thermal stability analysis, and numerical thermal boundary extraction to interpret the early thermal tendency in the (R_{isc}, U_{eq}) parameter plane.

1.4 Thesis Organization

The rest of this thesis is organized as follows. Chapter 2 establishes the early stage ISC induced T–SOC dynamical model and derives the time scale ratio $\epsilon(R_{isc}, U_{eq})$. The governing equations, heat generation terms, heat dissipation model, and time scale definitions are introduced in this chapter.

Chapter 3 presents the main dynamical analysis in the (R_{isc}, U_{eq}) parameter plane. The parameter space is partitioned according to the time scale ratio, and different analytical criteria are discussed for different time scale regimes. The simulated 70°C and 120°C thermal boundaries are then compared with representative constant- ϵ curves to explain the geometric relation between time scale separation and thermal safety boundaries.

Chapter 4 investigates the sensitivity of the thermal boundary interpretation to ambient temperature and external current. By changing T_{amb} and I_{ext} , this chapter examines whether the time scale based interpretation remains useful under different operating conditions, and how the warning and high temperature regions shift in the parameter plane.

Chapter 5 summarizes the main findings of the thesis, discusses the limitations of the present early stage model, and suggests possible directions for future work, including improved reaction kinetics, more detailed post threshold thermal runaway modeling, and experimental validation.

2 Two dimensional T–SOC Dynamical Model and Time Scale Partition

This chapter establishes the early stage internal short circuit model used in this thesis and derives the time scale parameter for dynamical system analysis. The purpose of this model is to describe the coupled evolution of temperature and state of charge before the battery enters the irreversible high temperature reaction stage. Therefore, this thesis selects the battery state of charge and the lumped battery temperature as state variables, and builds a two dimensional ordinary differential equation model.

The structure of this chapter is as follows. First, we introduce the basic assumptions. Then, the SOC equation and the thermal balance equation are derived by considering external discharge ohmic heat, internal short circuit heat, early chemical side reaction heat, and equivalent heat dissipation. Based on the obtained 2D T–SOC dynamical model, the characteristic time scales of temperature response and SOC depletion are defined. Finally, a parameter dependent time scale ratio is obtained, and this ratio is used to divide the parameter space into different dynamical regions.

2.1 Basic Assumptions

This thesis focuses on the early stage of thermal runaway induced by internal short circuit in lithium ion batteries. At this stage, the internal short circuit has already appeared, but the battery has not yet entered the heavily thermal runaway stage. The temperature is still within the early warning range, and the main goal of the model is to judge whether the current operating condition has enough thermal stability margin. Therefore, the model established in this thesis is an early warning model, not a complete thermal runaway propagation model.

The following basic assumptions are adopted in this model.

First, the battery is represented by a lumped parameter thermal model. The temperature inside the cell is assumed to be spatially uniform and is described by the equivalent battery temperature $T(t)$. This assumption ignores the spatial temperature gradient and local hot spot distribution inside the cell. Although this simplification cannot describe the detailed internal temperature field, it is suitable for cases where the battery management system needs a fast safety judgment.

Second, different from an open circuit internal short circuit model, this thesis introduces a fixed external discharge current. The reference case is chosen as

$$I_{ext} = 5 \text{ A}. \quad (2)$$

For the 25Ah cell considered in this thesis, this corresponds to $0.2C$. Since real battery operation may involve much larger currents, the effect of I_{ext} is not assumed to be negligible. Instead, it is examined separately in Chapter 4 through a current sensitivity study, including a larger current case $I_{ext} = 25 \text{ A}$, corresponding to $1C$ for the same cell. Therefore, the 5A case is used as the reference operating condition, and the robustness of the interpretation with respect to larger external currents is checked later.

Third, the internal short circuit is described by an equivalent internal short circuit resistance R_{isc} . A smaller R_{isc} represents a more severe internal short circuit, and it leads to a larger short circuit current and stronger ohmic heat generation. In this model, R_{isc} is treated as a key parameter rather than a state variable. This means that the model does not focus on the formation mechanism of the internal short circuit itself. Instead, it focuses on the influence of a given internal short circuit severity on the thermal safety state of the battery.

Fourth, only early chemical side reactions are retained in the thermal model. This thesis considers the heat released by solid electrolyte interphase reactions and anode side reactions, while the full chain reactions in the later thermal runaway stage are not included. Since this thesis focuses on the early temperature range, according to the kinetic model of Feng et al.^[9], the variations of the SEI reactant concentration and the anode reactant concentration are relatively small, so they are treated as constants. Under this assumption, the chemical heat generation term mainly depends on temperature through Arrhenius type reaction rates. This simplification allows the chemical heat generation term to be expressed as a function of temperature only, which is important for the later derivation of thermal stability analysis.

Finally, the heat dissipation process is described by a linear equivalent heat dissipation model. In this thesis, the detailed heat transfer through the cell casing, air convection, and cooling structure is not modeled separately. Instead, the heat exchange between the battery and the environment is represented by an equivalent heat dissipation term.

2.2 Derivation of the Two Dimensional T–SOC Model Equations

Under the above modeling assumptions, this thesis selects the state of charge $S(t)$ and the equivalent battery temperature $T(t)$ as two state variables.

First, the SOC evolution is considered. In this thesis, the decrease in SOC is caused by two current components: one is the external discharge current I_{ext} , and the other is the internal short circuit current I_{isc} . This thesis uses the equivalent internal short circuit resistance R_{isc} to describe the short circuit severity, and uses $U_{OCV}(S)$ to represent the battery open circuit voltage. Before deriving the SOC equation, the relationship between open circuit voltage and SOC needs to be determined. The open circuit voltage function $U_{OCV}(S)$ used in this thesis is not obtained by new experimental measurements, but is fitted from digitized points taken from the OCV–SOC curve given in Feng et al.'s study on internal short circuit detection for large format lithium ion batteries. Fig. 9 in that paper gives the relationship between battery open circuit voltage and SOC. Therefore, this thesis uses it as the data source of $U_{OCV}(S)$, and adopts a polynomial function to fit the digitized data^[10]. The fitted open circuit voltage function is written as

$$U_{OCV}(S) = 3.131 + 2.062S - 4.145S^2 + 5.761S^3 - 2.622S^4. \quad (3)$$

where $S \in [0, 1]$ denotes the battery SOC. The internal short circuit current, internal short circuit heat generation, and SOC depletion rate in the following model are all calculated based on this $U_{OCV}(S)$ function.

The internal short circuit current is approximately written as

$$I_{isc}(S) = \frac{U_{OCV}(S)}{R_{cell} + R_{isc}}, \quad (4)$$

where R_{cell} is the cell internal resistance. Therefore, the rate of change of SOC can be written as

$$\frac{dS}{dt} = -\frac{1}{3600Q_{cap}} [I_{ext} + I_{isc}(S)], \quad (5)$$

where Q_{cap} is the battery capacity in Ah, and the factor 3600 is used for unit conversion between Ah and coulomb. Substituting Eq. (4) into Eq. (5) gives

$$\frac{dS}{dt} = -\frac{1}{3600Q_{cap}} \left[I_{ext} + \frac{U_{OCV}(S)}{R_{cell} + R_{isc}} \right]. \quad (6)$$

This equation shows that both external discharge and internal short circuit accelerate the decrease in SOC. A smaller R_{isc} leads to a larger internal short circuit current, and therefore makes SOC depletion faster.

Next, the temperature equation is derived. The change in battery temperature is jointly determined by heat generation terms and heat dissipation terms. This thesis considers four heat sources or losses: external discharge ohmic heat, internal short circuit heat generation, early chemical side reaction heat generation, and heat dissipation to the environment.

The external discharge current generates ohmic heat through the cell internal resistance, and its power is written as

$$Q_{ext} = I_{ext}^2 R_{cell}. \quad (7)$$

.

The internal short circuit heat generation adopts a Joule heat expression with an efficiency factor, written as

$$Q_{isc} = \zeta I_{isc}^2 (R_{cell} + R_{isc}). \quad (8)$$

where ζ is the internal short circuit heat release efficiency factor. It is used to correct the case where not all electrical energy is converted into battery temperature rise during internal short circuit, and $\zeta = 0.28$ is adopted^[11].

Substituting Eq. (4) into the above equation gives

$$Q_{isc}(S, R_{isc}) = \zeta \frac{U_{OCV}^2(S)}{R_{cell} + R_{isc}}. \quad (9)$$

It can be seen from Eq. (9) that internal short circuit heat generation is not only related to R_{isc} , but is also connected with SOC through $U_{OCV}(S)$. Therefore, the coupling between SOC and temperature is mainly reflected through the internal short circuit heat generation term.

The early chemical side reaction heat mainly comes from SEI related reactions and anode side reactions. The heat generation power of a single reaction z can be expressed as

$$Q_z = h_z m_z \frac{dc_z^d}{dt}, \quad (10)$$

where h_z is the reaction enthalpy, m_z is the mass of the corresponding reactant, and $\frac{dc_z^d}{dt}$ is the decomposition rate of the reactant. Its Arrhenius form is

$$\frac{dc_z^d}{dt} = A_z c_z^{n_{z,1}} (1 - c_z)^{n_{z,2}} \exp\left(-\frac{E_{a,z}}{R_0 T}\right) g_z, \quad (11)$$

where A_z is the pre exponential factor, $E_{a,z}$ is the activation energy, R_0 is the gas constant, and g_z is the correction term for a specific reaction.

This thesis only retains the SEI reaction and the anode side reaction that are more important in the early temperature range. For the SEI reaction, there is

$$\frac{dc_{sei}^d}{dt} = A_{sei} c_{sei} \exp\left(-\frac{E_{a,sei}}{R_0 T}\right), \quad (12)$$

where c_{sei} is the SEI related reactant concentration. For the anode side reaction, the model also contains a correction term related to SEI coverage^[9], so the anode reaction rate is written as

$$\frac{dc_{an}^d}{dt} = A_{an} c_{an} \exp\left(-\frac{E_{a,an}}{R_0 T}\right) g_{an}, \quad (13)$$

where

$$g_{an} = \exp\left(-\frac{c_{sei}}{c_{sei,0}^{ref}}\right). \quad (14)$$

This correction term represents the suppression effect of the SEI layer on the reaction between the anode and the electrolyte. The thicker the SEI layer is, the slower the anode side reaction becomes.

Since this thesis focuses on the early temperature range, according to the previous assumptions, the SEI reactant concentration and the anode reactant concentration vary only slightly. Therefore, c_{sei} , c_{an} , and g_{an} can be treated as constants. Based on this, the SEI heat generation and the anode side reaction heat generation can be directly written as

$$\begin{aligned} Q_{sei}(T) &= h_{sei} m_{sei} A_{sei} c_{sei} \exp\left(-\frac{E_{a,sei}}{R_0 T}\right), \\ Q_{an}(T) &= h_{an} m_{an} A_{an} c_{an} g_{an} \exp\left(-\frac{E_{a,an}}{R_0 T}\right). \end{aligned} \quad (15)$$

Therefore, the early chemical side reaction heat generation term adopted in this thesis is

$$Q_{chem}(T) = Q_{sei}(T) + Q_{an}(T) \quad (16)$$

After this treatment, the chemical heat generation term is mainly controlled by temperature, which is convenient for deriving the thermal direction stability criterion later.

The heat dissipation from the battery to the environment is described by a linear equivalent heat dissipation model:

$$Q_{diss} = U_{eq}(T - T_{amb}), \quad (17)$$

where T_{amb} is the ambient temperature and U_{eq} is the equivalent heat dissipation coefficient. In this model, U_{eq} is used to describe how strongly the battery releases heat to the surrounding environment. It does not separate each heat transfer process in detail. Instead, it gives an equivalent linear description of the combined heat transfer through the cell casing and the external cooling condition.

Following the idea of a lumped parameter battery thermal model^[12], this thesis assumes a spatially uniform cell temperature and writes the thermal balance as

$$m C_p \frac{dT}{dt} = Q_{ext} + Q_{isc} + Q_{chem} - Q_{diss}, \quad (18)$$

where m is the battery mass, and C_p is the specific heat capacity.

Finally, combining Eq. (6) and Eq. (18) gives the two dimensional T-SOC dynamical model adopted in this thesis:

$$\begin{cases} \dot{S} = -\frac{1}{3600Q_{cap}} \left[I_{ext} + \frac{U_{OCV}(S)}{R_{cell} + R_{isc}} \right], \\ \dot{T} = \frac{1}{mC_p} \left[I_{ext}^2 R_{cell} + \zeta \frac{U_{OCV}^2(S)}{R_{cell} + R_{isc}} + Q_{chem}(T) - U_{eq}(T - T_{amb}) \right]. \end{cases} \quad (19)$$

In this model, R_{isc} mainly affects the internal short circuit current, SOC depletion rate, and internal short circuit heat generation intensity. U_{eq} directly affects the heat dissipation capability of the battery. Therefore, R_{isc} and U_{eq} are two key parameters in the later time scale partition and stability analysis.

2.3 Model Parameters

The parameters used in the two dimensional T-SOC model are summarized in this subsection. In this thesis, the internal short circuit resistance R_{isc} and the equivalent heat dissipation coefficient U_{eq} are selected as the two main control variables. The remaining parameters are treated as constant values.

Table 1: Electrical and thermal parameters used in the T-SOC model

Symbol	Meaning	Value	Unit
Q_{cap}	Battery capacity	25	Ah
I_{ext}	Reference external discharge current	5	A
R_{cell}	Cell internal resistance	2×10^{-3}	Ω
m	Battery mass	0.72	kg
C_p	Specific heat capacity	1100	J/(kg·K)
T_{amb}	Ambient temperature	313.15	K
ζ	Internal short circuit heat release efficiency factor	0.28	1

The parameters in this table are taken from Ref. 9.

The early chemical heat generation model only includes the SEI reaction and the anode reaction. The corresponding kinetic parameters are listed in Table 2.

Table 2: Parameters of early chemical side reactions

Symbol	Meaning	Value	Unit
R_0	Gas constant	8.314	J/(mol·K)
h_{sei}	Reaction enthalpy of SEI reaction	150	J/g
m_{sei}	Mass of SEI related reactants	100.58	g
c_{sei}	Normalized SEI reactant concentration	0.15	1
A_{sei}	Pre exponential factor of SEI reaction	1.667×10^{15}	s^{-1}
$E_{a,sei}$	Activation energy of SEI reaction	1.3508×10^5	J/mol
$T_{onset,sei}$	Onset temperature of SEI reaction	323.15	K
$n_{sei,1}$	Reaction order of SEI reaction	1	1
$n_{sei,2}$	Reaction order of SEI reaction	0	1
h_{an}	Reaction enthalpy of anode side reaction	1714	J/g
m_{an}	Mass of anode related reactants	100.58	g
c_{an}	Normalized anode reactant concentration	1	1
A_{an}	Pre exponential factor of anode side reaction	0.038	s^{-1}
$E_{a,an}$	Activation energy of anode side reaction	3.3×10^4	J/mol
$T_{onset,an}$	Onset temperature of anode side reaction	323.15	K
$n_{an,1}$	Reaction order of anode side reaction	1	1
$n_{an,2}$	Reaction order of anode side reaction	0	1
$c_{sei,0}^{ref}$	Reference SEI concentration in g_{an}	0.15	1
g_{an}	Correction factor of anode side reaction	$\exp(-c_{sei}/c_{sei,0}^{ref})$	1

The chemical kinetic parameters in this table are taken from Ref. 9.

In addition to the fixed parameters listed above, R_{isc} and U_{eq} are treated as the two main scanning parameters in this thesis. The internal short circuit resistance is varied over a logarithmic range,

$$R_{isc} \in [0.01, 10] \Omega, \quad (20)$$

while the equivalent heat dissipation coefficient is varied over the range

$$U_{eq} \in [0.05, 10] \text{ W/K}. \quad (21)$$

The logarithmic range of R_{isc} is used because the internal short circuit severity can change over several orders of magnitude.

These two parameters define the (R_{isc}, U_{eq}) parameter plane used in the time scale partition, thermal stability analysis, and numerical boundary extraction in the following chapters. All other parameters in Tables 1 and 2 are kept fixed unless otherwise stated.

2.4 Time Scale Parameter and Parameter Space Partition

After obtaining the two dimensional T-SOC dynamical model, this thesis further constructs a time scale parameter to determine whether the system has a clear fast slow structure under given R_{isc} and U_{eq} . Since temperature and SOC evolve at the same time in the model, different mathematical analysis should be used in different time scale regions. If the time scales of temperature and SOC are clearly separated, the system can be analyzed using fast slow dynamical system methods^[13]. If the two time scales are close to each other, other more conservative stability analysis are needed.

First, the characteristic time scale of the temperature response is considered. From the temperature equation, the linear heat dissipation term is

$$-U_{eq}(T - T_{amb}). \quad (22)$$

In this thesis, the thermal response time scale is based on the heat capacity and the equivalent heat dissipation coefficient:

$$\tau_T(U_{eq}) = \frac{mC_p}{U_{eq}}. \quad (23)$$

Here, mC_p is the total heat capacity of the battery, and U_{eq} is the equivalent heat dissipation coefficient. Therefore, a larger U_{eq} leads to a shorter thermal response time scale.

It should be emphasized that τ_T in Eq. (23) is a nominal thermal response time scale. It is not the actual time needed for the cell temperature to reach a prescribed warning threshold. Instead, it is obtained from the linear heat dissipation part of the thermal equation. The chemical heat generation term $Q_{chem}(T)$ is not included in this nominal time scale definition because it is a nonlinear state dependent heat source. Its influence on the local thermal direction is controlled by the temperature derivative $Q'_{chem}(T)$, not by a constant coefficient.

This also means that, if $Q_{chem}(T)$ were included directly in the time scale definition, the resulting quantity would no longer depend only on the parameters R_{isc} and U_{eq} . It would also depend on the current temperature state and would change along each trajectory. Such a state dependent quantity is useful for local stability analysis, but it cannot be represented as a fixed partition in the (R_{isc}, U_{eq}) parameter plane. Since the purpose of this thesis is to obtain a parameter-plane based early risk indication from R_{isc} and U_{eq} under a prescribed operating condition, the time scale ratio is defined as a nominal parameter level quantity.

In the lower temperature range, especially below 70°C, $Q'_{chem}(T)$ is much smaller than the representative cooling coefficients. This is also the temperature range reached in the strongly separated region considered later, where the simulated maximum temperature remains below 70°C. Therefore, in the region where the fast slow separation is actually used, the influence of chemical heat on the nominal thermal response time is weak compared with the cooling coefficient. This supports using the cooling based time scale mC_p/U_{eq} for the parameter level definition of $\epsilon(R_{isc}, U_{eq})$. Figure 1 illustrates this point.

As the temperature approaches the high temperature range, $Q'_{chem}(T)$ increases rapidly and can become comparable to or even larger than weak cooling coefficients. This shows why $Q_{chem}(T)$ must still be retained in the full model and treated through the later local thermal stability analysis and numerical simulations. Thus, omitting $Q_{chem}(T)$ from τ_T only means that $\epsilon(R_{isc}, U_{eq})$ is used as a nominal parameter level indicator. It does not mean that chemical heat is neglected in the thermal model.

Next, consider the characteristic time scale of SOC depletion. According to Eq. (6), the decrease in SOC is caused by both the external discharge current and the internal short circuit current. To obtain a parameter dependent estimate, a reference open circuit voltage is introduced as

$$U_{ref} = \max_{S \in [0,1]} U_{OCV}(S). \quad (24)$$

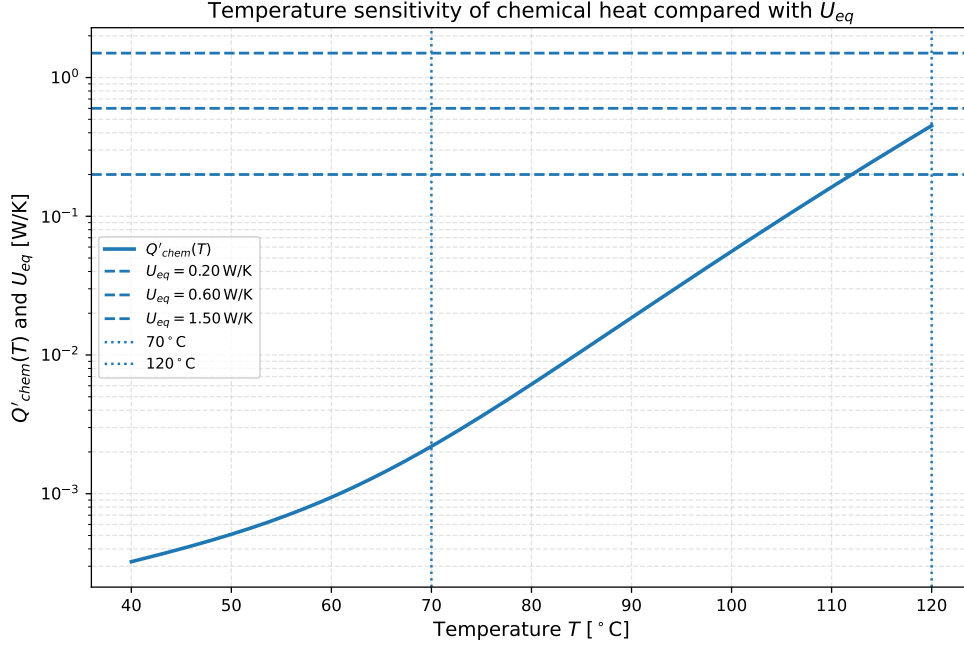


Figure 1: Comparison of the temperature sensitivity of chemical heat generation $Q'_{chem}(T)$ with representative cooling coefficients U_{eq} . In the lower temperature range, $Q'_{chem}(T)$ is much smaller than the representative cooling coefficients. Near the high temperature range, $Q'_{chem}(T)$ increases rapidly and can become comparable to weak cooling.

Then the reference total discharge current is written as

$$I_{ref}(R_{isc}) = I_{ext} + \frac{U_{ref}}{R_{cell} + R_{isc}}. \quad (25)$$

The first term represents the fixed external discharge current, and the second term represents the reference current caused by internal short circuit. Based on this reference current, the SOC depletion time scale is defined as

$$\tau_S(R_{isc}) = \frac{3600Q_{cap}}{I_{ref}(R_{isc})}. \quad (26)$$

Based on the thermal response time scale and the SOC depletion time scale, the time scale ratio is defined as

$$\epsilon(R_{isc}, U_{eq}) = \frac{\tau_T(U_{eq})}{\tau_S(R_{isc})}. \quad (27)$$

Substituting Eq. (23) and Eq. (26) into Eq. (27) gives

$$\epsilon(R_{isc}, U_{eq}) = \frac{mC_p}{3600Q_{cap}U_{eq}} \left[I_{ext} + \frac{U_{ref}}{R_{cell} + R_{isc}} \right]. \quad (28)$$

Eq. (28) shows that ϵ describes the relative size between the nominal thermal response time scale and the SOC depletion time scale. When ϵ is small, the cooling based thermal response time is short compared with the SOC depletion time, and the system is closer to a fast slow system. When ϵ is large, temperature and SOC evolve on comparable time scales, and the system no longer has a clear fast slow structure. With this definition, $\epsilon(R_{isc}, U_{eq})$ should be interpreted as a nominal structural indicator rather than an exact local eigenvalue. The nonlinear chemical heat term is accounted for later through the local thermal attraction condition, the Jacobian based indicator $\lambda_2(T)$, and the full numerical simulations.

Furthermore, from Eq. (28), when other parameters are fixed,

$$\frac{\partial \epsilon}{\partial U_{eq}} = -\frac{mC_p}{3600Q_{cap}U_{eq}^2} \left[I_{ext} + \frac{U_{ref}}{R_{cell} + R_{isc}} \right] < 0, \quad (29)$$

and

$$\frac{\partial \epsilon}{\partial R_{isc}} = -\frac{mC_p U_{ref}}{3600 Q_{cap} U_{eq} (R_{cell} + R_{isc})^2} < 0. \quad (30)$$

Therefore, increasing U_{eq} shortens the nominal thermal response time scale and decreases ϵ . Increasing R_{isc} weakens the internal short circuit current and slows SOC depletion, which also decreases ϵ . It should also be noted that because a fixed external discharge current I_{ext} is considered in this thesis, ϵ does not approach zero when R_{isc} becomes very large.

According to the value of $\epsilon(R_{isc}, U_{eq})$, the parameter space is divided into three regions:

$$\epsilon(R_{isc}, U_{eq}) < 0.1, \quad (31)$$

which corresponds to the strong fast slow separation region. In this region, the nominal thermal response time is much shorter than the SOC depletion time, so geometric singular perturbation theory (GSPT) can be used to analyze the critical manifold and fast thermal attraction of the temperature dynamics.

$$0.1 \leq \epsilon(R_{isc}, U_{eq}) < 0.3, \quad (32)$$

which corresponds to the weak fast slow separation region. In this region, the system still has some time scale difference, but the separation is not strong enough for strict GSPT analysis. Therefore, a Jacobian based local thermal direction stability criterion will be used to analyze whether small temperature perturbations are locally damped or amplified.

$$\epsilon(R_{isc}, U_{eq}) \geq 0.3, \quad (33)$$

which corresponds to the non fast slow separation region. In this region, temperature and SOC evolve on similar time scales, so the system is not suitable to be simplified as a standard fast slow system. The full numerical thermal boundary comparison is therefore used as a conservative risk indication in this region.

From Eq. (28), the boundary curve for a given $\epsilon = \epsilon_0$ can be obtained as

$$U_{eq}^{(\epsilon_0)}(R_{isc}) = \frac{mC_p}{3600 Q_{cap} \epsilon_0} \left[I_{ext} + \frac{U_{ref}}{R_{cell} + R_{isc}} \right]. \quad (34)$$

Therefore, the two curves $\epsilon = 0.1$ and $\epsilon = 0.3$ can be used as the boundaries between the strong fast slow separation region, the weak fast slow separation region, and the non fast slow separation region.

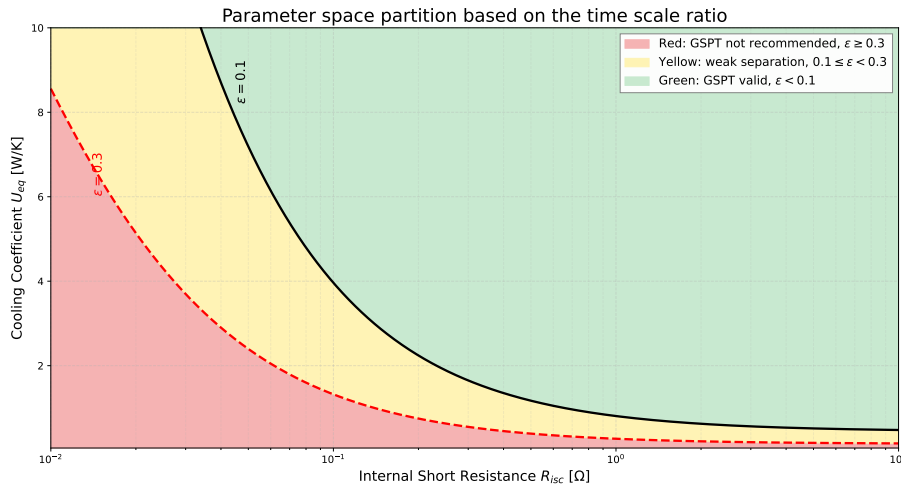


Figure 2: Parameter space partition based on the time scale ratio $\epsilon(R_{isc}, U_{eq})$.

As shown in Fig. 2, the parameter space can be divided into three dynamical regions according to the time scale ratio. Larger U_{eq} and larger R_{isc} usually correspond to smaller ϵ , so the system is more likely to show a fast slow structure. Smaller U_{eq} or smaller R_{isc} increases ϵ , and the system gradually enters the weak fast slow or non fast slow separation region. This partition provides the basis for selecting different mathematical analysis in the following chapters.

3 Stability Analysis of Different Time Scale Regions

In Chapter 2, a two dimensional T-SOC dynamical model was established for early stage internal short circuit under a fixed external discharge current. Based on the characteristic time scales of temperature response and SOC depletion, the parameter space was divided into three regions according to the time scale ratio $\epsilon(R_{isc}, U_{eq})$.

This chapter analyzes the thermal stability of these three regions. The purpose of this chapter is not to apply one single method to all operating conditions. Instead, different analytical tools are selected according to the time scale structure of the system. This provides a more flexible and interpretable framework for early stage thermal safety analysis.

3.1 Strong Fast Slow Separation Region

This subsection focuses on the strong fast slow separation region defined in Chapter 2, where the temperature evolves much faster than the SOC. In this region, the temperature T is treated as the fast variable, while the SOC is treated as a slow variable.

3.1.1 Fast subsystem and critical manifold

From the two dimensional T-SOC model, the temperature equation can be written as

$$\dot{T} = \frac{1}{mC_p} \Phi(T; S, R_{isc}, U_{eq}), \quad (35)$$

where

$$\Phi(T; S, R_{isc}, U_{eq}) = I_{ext}^2 R_{cell} + \zeta \frac{U_{OCV}^2(S)}{R_{cell} + R_{isc}} + Q_{chem}(T) - U_{eq}(T - T_{amb}). \quad (36)$$

Here, T is the fast variable, while S , R_{isc} , and U_{eq} are treated as parameters in the fast subsystem. Since S changes much more slowly than T on the fast time scale, S is frozen during the fast thermal motion, and the fast subsystem is written as

$$\frac{dT}{dt} = \frac{1}{mC_p} \Phi(T; S, R_{isc}, U_{eq}), \quad \frac{dS}{dt} = 0. \quad (37)$$

The critical manifold is the set of equilibrium points of the fast subsystem. It is defined by

$$C_0 = \{(S, T) : \Phi(T; S, R_{isc}, U_{eq}) = 0\}. \quad (38)$$

This equation has a clear physical meaning. On the critical manifold, the external discharge heat, the internal short circuit heat, and the early chemical heat are exactly balanced by the heat dissipation to the environment. The critical manifold is introduced because it describes the quasi steady thermal balance reached by the fast temperature dynamics for each frozen SOC value. It provides the geometric object on which the fast thermal attraction and the reduced slow evolution can be interpreted.

Although S is frozen in the fast subsystem, it is still in the critical manifold C_0 . This is because different values of S give different internal short circuit heat through $U_{OCV}(S)$. By collecting the fast thermal equilibria for all admissible $S \in [0, 1]$, the critical manifold is obtained in the (S, T) phase plane.

Since the critical manifold depends on the parameter pair (R_{isc}, U_{eq}) , it is shown here under one representative parameter condition in the strong fast slow separation region, together with the vector field and trajectories from different initial conditions.

As shown in Fig. 3, the initial states are not necessarily located on the critical manifold. The vector field indicates the local direction of the full (S, T) dynamics in the phase plane. Around the critical manifold, the vectors point toward the thermal balance curve from both higher and lower temperature sides, which illustrates the attracting character of this branch. The trajectories starting from different initial conditions first move rapidly in the temperature direction and approach C_0 . After they get close to C_0 , their motion becomes mainly guided by the slower SOC evolution. This illustrates why the critical manifold is the basic geometric object for the GSPT analysis in the strong fast slow separation region. The geometric meaning of this result is consistent with the fast slow interpretation. In the singular limit $\epsilon = 0$, the reduced slow motion takes place on the critical manifold C_0 . For the real system with small positive ϵ , the trajectory does not lie exactly on C_0 . By Fenichel theory^[14], if the considered part of C_0 is normally hyperbolic and attracting, then there exists a nearby attracting slow manifold C_ϵ , which is $O(\epsilon)$ close to C_0 . The full trajectory therefore first approaches this nearby slow manifold rapidly in the temperature direction, and then evolves slowly close to it as SOC changes.

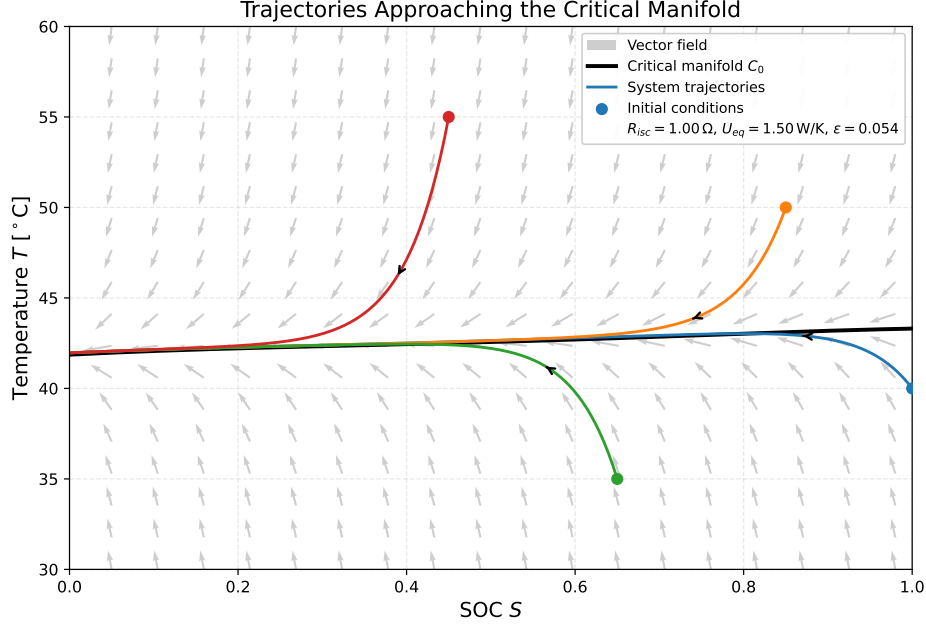


Figure 3: Representative critical manifold, vector field, and system trajectories from different initial conditions under a fixed parameter pair (R_{isc}, U_{eq}) .

3.1.2 Fast thermal attraction and temperature check

The stability of the critical manifold in the fast thermal direction is determined by the derivative of the fast subsystem with respect to T . Since the thermal equation can be written as

$$\dot{T} = \frac{1}{mC_p} \Phi(T, S; R_{isc}, U_{eq}), \quad (39)$$

and $1/(mC_p) > 0$, the sign of the fast derivative is determined by

$$\frac{\partial \Phi}{\partial T} = Q'_{chem}(T) - U_{eq}. \quad (40)$$

The external discharge heat and the internal short circuit heat do not directly depend on T , so they do not appear in this derivative.

On the critical manifold

$$C_0 = \{(S, T) : \Phi(T; S, R_{isc}, U_{eq}) = 0\}, \quad (41)$$

if

$$Q'_{chem}(T) < U_{eq}, \quad (42)$$

then the fast thermal direction is attracting. A small temperature perturbation decays back toward the critical manifold. If

$$Q'_{chem}(T) > U_{eq}, \quad (43)$$

then the fast thermal direction is repelling. A small temperature perturbation is amplified.

The sign changes when

$$Q'_{chem}(T) = U_{eq}. \quad (44)$$

This equation gives the temperature at which the fast thermal attraction changes sign. Below this temperature, the chemical heat generation increases slowly enough that the heat dissipation term can still damp a small temperature perturbation. Above this temperature, the temperature sensitivity of the chemical heat generation becomes stronger than the cooling effect, and a small temperature perturbation tends to grow. Therefore, in this thesis, Eq. (40) is used as a local thermal attraction check within the early stage temperature range. It provides a simple criterion for judging whether the temperature direction is locally attracting or locally amplifying, without requiring a full equilibrium analysis of the two dimensional system.

To use Eq. (44) as a temperature boundary, it is useful to show that $Q'_{chem}(T)$ is monotone in the early temperature range considered in this thesis. This ensures that, for each fixed U_{eq} , the equation $Q'_{chem}(T) = U_{eq}$ defines at most one critical temperature.

From the chemical heat model in Chapter 2, the total early chemical heat generation is

$$Q_{chem}(T) = q_{sei}(T) + q_{an}(T), \quad (45)$$

where

$$q_{sei}(T) = h_{sei} m_{sei} A_{sei} c_{sei} \exp\left(-\frac{E_{a,sei}}{R_0 T}\right), \quad (46)$$

and

$$q_{an}(T) = h_{an} m_{an} A_{an} c_{an} g_{an} \exp\left(-\frac{E_{a,an}}{R_0 T}\right). \quad (47)$$

Here, c_{sei} , c_{an} , and g_{an} are treated as constants in the early stage model. Thus, the two heat generation terms have the same Arrhenius structure. For compact notation, define

$$A_{sei}^* = h_{sei} m_{sei} A_{sei} c_{sei} > 0, \quad A_{an}^* = h_{an} m_{an} A_{an} c_{an} g_{an} > 0. \quad (48)$$

Then

$$Q_{chem}(T) = A_{sei}^* \exp\left(-\frac{E_{a,sei}}{R_0 T}\right) + A_{an}^* \exp\left(-\frac{E_{a,an}}{R_0 T}\right). \quad (49)$$

Taking the first derivative gives

$$Q'_{chem}(T) = A_{sei}^* \exp\left(-\frac{E_{a,sei}}{R_0 T}\right) \frac{E_{a,sei}}{R_0 T^2} + A_{an}^* \exp\left(-\frac{E_{a,an}}{R_0 T}\right) \frac{E_{a,an}}{R_0 T^2}. \quad (50)$$

Taking one more derivative, the second derivative is

$$\begin{aligned} Q''_{chem}(T) &= A_{sei}^* \frac{E_{a,sei}}{R_0 T^3} \exp\left(-\frac{E_{a,sei}}{R_0 T}\right) \left(\frac{E_{a,sei}}{R_0 T} - 2\right) \\ &\quad + A_{an}^* \frac{E_{a,an}}{R_0 T^3} \exp\left(-\frac{E_{a,an}}{R_0 T}\right) \left(\frac{E_{a,an}}{R_0 T} - 2\right). \end{aligned} \quad (51)$$

All prefactors and exponential terms in Eq. (51) are positive. Therefore, both terms in $Q''_{chem}(T)$ are positive if

$$T < \min\left(\frac{E_{a,sei}}{2R_0}, \frac{E_{a,an}}{2R_0}\right). \quad (52)$$

Using the kinetic parameters adopted in this thesis,

$$\frac{E_{a,sei}}{2R_0} \approx 8124 \text{ K}, \quad \frac{E_{a,an}}{2R_0} \approx 1985 \text{ K}. \quad (53)$$

The smaller value is still far above the early temperature range considered in this model. Hence, in the physical temperature range of interest,

$$Q''_{chem}(T) > 0. \quad (54)$$

This proves that $Q'_{chem}(T)$ is strictly increasing with temperature in the early stage temperature range. Consequently, the fast derivative

$$\Phi_T(T; S, R_{isc}, U_{eq}) = Q'_{chem}(T) - U_{eq} \quad (55)$$

is also strictly increasing with respect to T .

For a given cooling coefficient U_{eq} , the critical temperature for fast thermal attraction is defined by

$$Q'_{chem}(T_{crit}(U_{eq})) = U_{eq}. \quad (56)$$

Because $Q'_{chem}(T)$ is strictly increasing, this critical temperature is unique if it exists in the considered temperature range. For temperatures below this boundary,

$$T < T_{crit}(U_{eq}) \quad \Rightarrow \quad \Phi_T < 0, \quad (57)$$

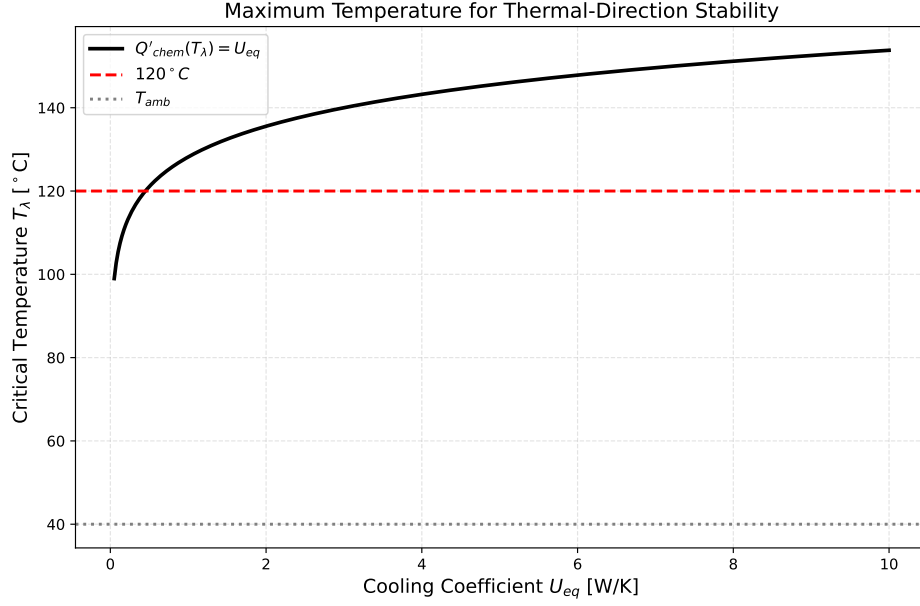


Figure 4: Critical temperature for fast thermal attraction defined by $Q'_{chem}(T_{crit}) = U_{eq}$.

and the fast thermal direction is attracting. For temperatures above this boundary,

$$T > T_{crit}(U_{eq}) \Rightarrow \Phi_T > 0, \quad (58)$$

and fast thermal attraction is lost.

Figure 4 shows the relation between U_{eq} and T_{crit} . A larger cooling coefficient increases the critical temperature. This means that stronger heat dissipation keeps the fast thermal direction attracting up to a higher temperature. The boundary does not directly depend on R_{isc} , because the electrical heat term does not explicitly depend on T in the present model and therefore disappears after taking the derivative with respect to T . However, this does not mean that R_{isc} is unimportant. The internal short circuit resistance still affects the electrical heat generation, the location of the critical manifold, and the actual temperature trajectory.

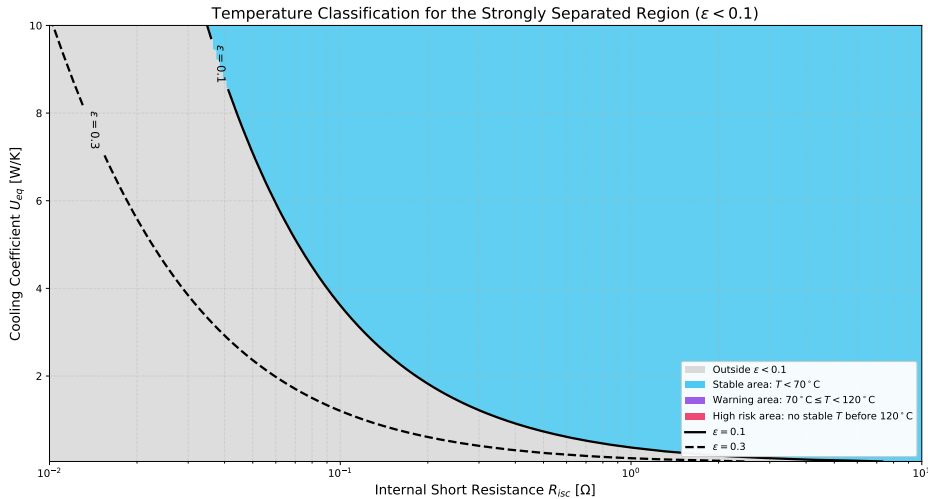


Figure 5: Temperature check inside the strongly separated region $\epsilon < 0.1$.

This interpretation should be understood within the early stage temperature range considered in this thesis. The model is not intended to describe the severe thermal runaway process after the temperature exceeds 120°C. Therefore, the part of the curve above 120°C is used only as a mathematical continuation of the local attraction boundary. It should not be interpreted as a physical prediction of the post runaway dynamics.

This boundary also gives a simple real time check for a battery management system. If the current cell

temperature $T_{real}(t)$ and the effective cooling coefficient U_{eq} are available, the sign of

$$Q'_{chem}(T_{real}(t)) - U_{eq} \quad (59)$$

can be evaluated directly. If this value is negative, the current state is still in the fast thermal attracting region. If this value is nonnegative, the current temperature has reached the local attraction boundary or entered the region where fast thermal attraction is lost.

In this thesis, real time BMS temperature measurements are not available. Therefore, simulated temperatures are used only as an example to show how the mathematically derived boundary works for the full nonlinear model.

Applying this check to the strongly separated region $\epsilon < 0.1$ gives the result shown in Fig. 5. For the present parameter choice, the simulated temperatures in this region remain below the fast thermal attraction boundary and also below the BMS warning threshold 70°C . Therefore, the whole strongly separated region is classified as stable under the present analysis.

3.2 Jacobian-Based Local Thermal Direction Stability

For $\epsilon \geq 0.1$, the time scale separation is no longer strong enough to rely on a strict GSPT description. However, the local thermal growth rate of the original two dimensional system can still be used to check whether temperature perturbations decay within the conservative reachable temperature range.

It should be emphasized first that the following analysis is not an equilibrium point stability analysis. In the present model, SOC generally keeps decreasing, so the full system usually does not have a true equilibrium point. The local linearization is only used to describe whether a small temperature perturbation around a state point (S, T) tends to decay or grow.

The T-SOC system can be written as

$$\dot{S} = f(S, T), \quad \dot{T} = g(S, T), \quad (60)$$

where

$$f(S, T) = -\frac{1}{3600Q_{cap}} \left[I_{ext} + \frac{U_{OCV}(S)}{R_{cell} + R_{isc}} \right], \quad (61)$$

and

$$g(S, T) = \frac{1}{mC_p} \left[I_{ext}^2 R_{cell} + \zeta \frac{U_{OCV}^2(S)}{R_{cell} + R_{isc}} + Q_{chem}(T) - U_{eq}(T - T_{amb}) \right]. \quad (62)$$

Since $f(S, T)$ does not depend on T , the Jacobian matrix along a state point (S, T) has a lower triangular form:

$$J(S, T) = \begin{bmatrix} f_S & 0 \\ g_S & g_T \end{bmatrix}. \quad (63)$$

Therefore, the two local growth rates are

$$\lambda_1 = f_S, \quad \lambda_2 = g_T. \quad (64)$$

More specifically,

$$\lambda_1 = -\frac{1}{3600Q_{cap}} \frac{U'_{OCV}(S)}{R_{cell} + R_{isc}}, \quad (65)$$

and

$$\lambda_2(T) = \frac{1}{mC_p} [Q'_{chem}(T) - U_{eq}]. \quad (66)$$

The value of λ_2 describes the local growth rate in the thermal direction. If

$$\lambda_2(T) < 0, \quad (67)$$

then a small temperature perturbation tends to decay. If

$$\lambda_2(T) > 0, \quad (68)$$

then a small temperature perturbation tends to grow. Thus, the local thermal direction stability condition is

$$Q'_{chem}(T) < U_{eq}. \quad (69)$$

It is useful to note that the expression of λ_2 has the same mathematical form as the fast derivative Φ_T used in Section 3.1.2, up to the positive factor $1/(mC_p)$. However, the interpretation is different. In Section 3.1.2, $\Phi_T < 0$ is used to identify the attracting branch of the critical manifold in the GSPT framework. Here, $\lambda_2 < 0$ is used to describe local decay of a temperature perturbation in the original two dimensional system.

The physical meaning of a temperature perturbation should be understood locally. It refers to a small deviation of the cell temperature from the trajectory, for example due to sensor error, internal-surface temperature mismatch, or model prediction error. Therefore, $\lambda_2(T) < 0$ does not directly mean that thermal runaway is impossible. Rather, it means that at the current temperature state, a small thermal deviation tends to decay. This makes λ_2 useful as a BMS real time indicator: if the current temperature $T(t)$ and the effective cooling coefficient U_{eq} are available, the sign of $\lambda_2(T(t))$ can be evaluated directly.

To illustrate this idea, three representative cases are shown below. In each case, the upper picture shows the vector field of the full T-SOC system in the (S, T) phase plane, together with the simulated trajectory starting from the initial condition $(S_0, T_0) = (1, 40^\circ\text{C})$. The arrows indicate the local direction of the dynamics, while the trajectory shows how the state moves as SOC decreases and temperature evolves. The lower picture shows the corresponding value of $\lambda_2(T(t))$ along the same trajectory. The shaded regions in the lower picture indicate the intervals where the thermal direction is locally damping or locally amplifying.

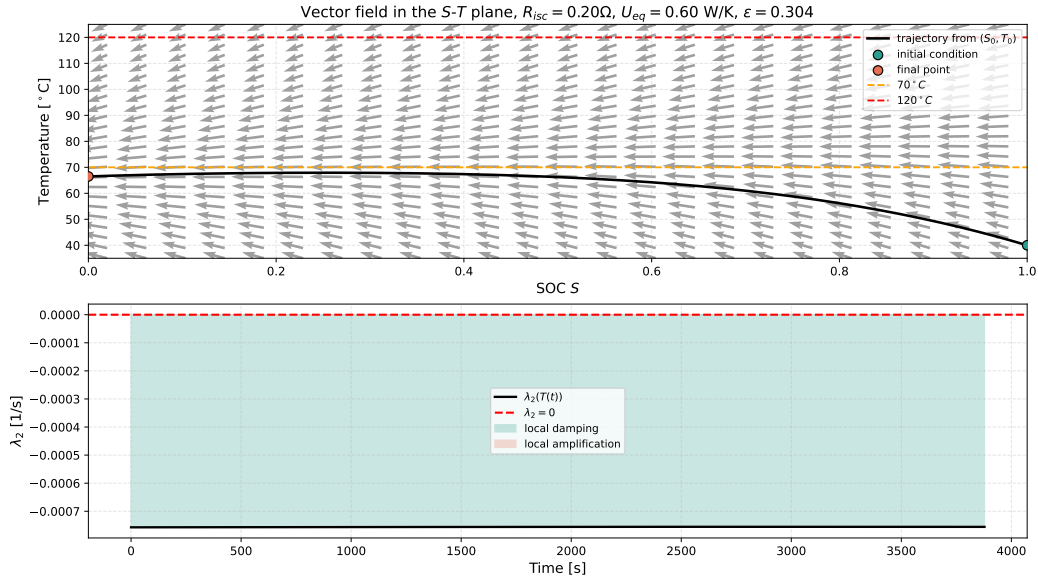


Figure 6: Representative phase-plane trajectory and local thermal stability for $R_{isc} = 0.20 \Omega$, $U_{eq} = 0.60 \text{ W/K}$, and $\epsilon = 0.304$. The upper picture shows the vector field in the (S, T) plane and the simulated trajectory from the initial condition. The lower picture shows $\lambda_2(T(t))$, which remains negative throughout the trajectory. The trajectory stays below 70°C .

In Fig. 6, the phase-plane trajectory remains below 70°C , and $\lambda_2(T(t))$ stays negative during the whole simulation. The vector field shows that the trajectory moves toward lower SOC without entering the warning temperature range. This represents a low temperature stable case, the cell remains below the BMS warning threshold, and small thermal perturbations are locally damped.

In Fig. 7, the phase-plane trajectory crosses the 70°C warning level, but it remains below 120°C . Although the cell enters the warning temperature range, $\lambda_2(T(t))$ stays negative along the whole trajectory. This means that the temperature level is no longer fully safe from a BMS viewpoint, but the thermal direction is still locally damping in the present model.

In Fig. 8, the cooling coefficient is kept the same as in Fig. 7, while the internal short resistance is reduced. The vector field and the simulated trajectory show that the stronger short circuit drives the state toward a higher temperature region as SOC decreases. Before the trajectory reaches 120°C , $\lambda_2(T(t))$ crosses zero and becomes positive. Thus, the system enters a locally amplifying thermal direction before the early thermal runaway boundary is reached. This example shows why a fixed temperature threshold alone is not sufficient to describe the local thermal stability of the trajectory.

These representative cases clarify the role of the Jacobian analysis. The temperature thresholds 70°C and 120°C describe the severity of the reached temperature level, while $\lambda_2(T(t))$ describes whether small thermal

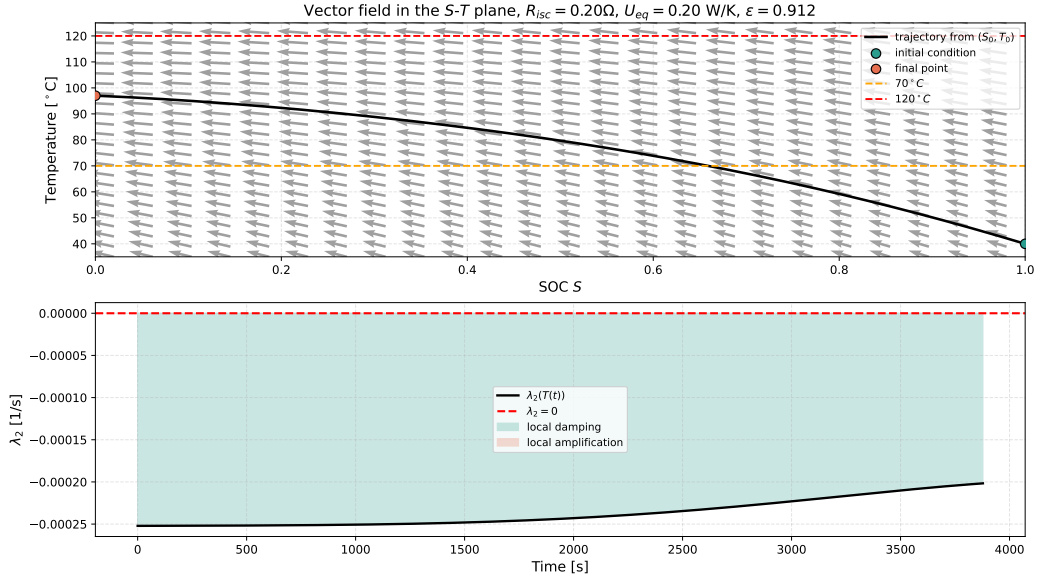


Figure 7: Representative phase-plane trajectory and local thermal stability for $R_{isc} = 0.20\Omega$, $U_{eq} = 0.20$ W/K, and $\epsilon = 0.912$. The upper picture shows the vector field in the (S, T) plane and the simulated trajectory. The trajectory crosses the 70°C warning level but remains below 120°C . The lower picture shows that $\lambda_2(T(t)) < 0$ throughout the trajectory.

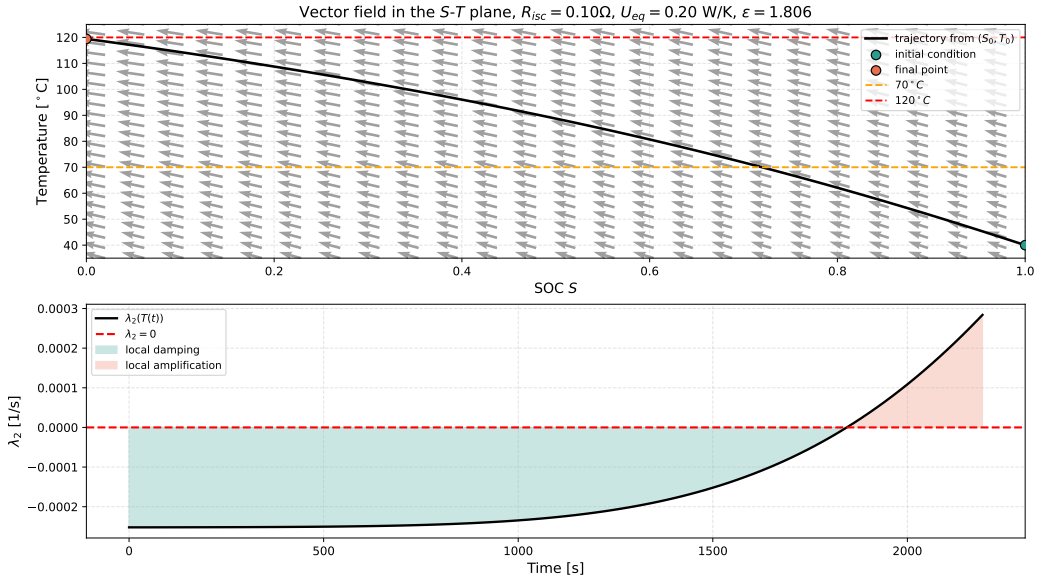


Figure 8: Representative phase-plane trajectory and local thermal stability for $R_{isc} = 0.10\Omega$, $U_{eq} = 0.20$ W/K, and $\epsilon = 1.806$. Compared with the previous case, the stronger internal short circuit drives the trajectory toward a higher temperature region in the (S, T) plane. The lower picture shows that $\lambda_2(T(t))$ becomes positive before the trajectory reaches 120°C .

deviations are locally damped or amplified along the trajectory. The vector field gives a phase plane view of the full nonlinear T-SOC dynamics, while the Jacobian based indicator provides a local stability interpretation along the simulated trajectory.

3.3 Time-scale structure of simulated thermal boundaries

The Jacobian indicator in Section 3.2 is a state based criterion. If the current temperature $T(t)$ and the effective cooling coefficient U_{eq} are available, the sign of $\lambda_2(T(t))$ can be evaluated directly to determine whether the thermal direction is locally damping or amplifying. However, this criterion still requires temperature information. It does not give a parameter space boundary in the (R_{isc}, U_{eq}) plane.

For this reason, this subsection considers a different question. Instead of asking whether the current thermal state is locally stable, we ask whether a given parameter pair (R_{isc}, U_{eq}) tends to drive the full two dimensional ode system toward a high-temperature regime. It only uses the short circuit resistance and the cooling coefficient, and therefore it is different from the real time state check in Section 3.2.

3.3.1 Structural relation between thermal boundaries and time scale parameter

Before presenting the simulation result, it is useful to first examine why a relation between thermal boundaries and the time scale parameter may be expected from the model structure.

Let

$$x = R_{cell} + R_{isc}, \quad K = \frac{mC_p}{3600Q_{cap}}. \quad (70)$$

From Chapter 2, the time scale ratio is

$$\epsilon(R_{isc}, U_{eq}) = \frac{K}{U_{eq}} \left[I_{ext} + \frac{U_{ref}}{x} \right]. \quad (71)$$

For a fixed value $\epsilon = \epsilon_c$, Eq. (71) gives

$$U_{eq}^{\epsilon_c}(R_{isc}) = \frac{KI_{ext}}{\epsilon_c} + \frac{KU_{ref}}{\epsilon_c} \frac{1}{R_{cell} + R_{isc}}. \quad (72)$$

Thus, in the (R_{isc}, U_{eq}) parameter plane, each constant ϵ curve can be written in the form

$$U_{eq}^{\epsilon_c} = A_\epsilon + B_\epsilon \frac{1}{R_{cell} + R_{isc}}, \quad (73)$$

where

$$A_\epsilon = \frac{KI_{ext}}{\epsilon_c}, \quad B_\epsilon = \frac{KU_{ref}}{\epsilon_c}. \quad (74)$$

A similar structure appears in the thermal boundary. Consider a fixed temperature level T_b , and define

$$\Delta T_b = T_b - T_{amb}. \quad (75)$$

Near a temperature boundary, the balance between heat generation and heat dissipation can be approximated by

$$I_{ext}^2 R_{cell} + \zeta \frac{U_{OCV}^2(S_b)}{R_{cell} + R_{isc}} + Q_{chem}(T_b) = U_{eq} \Delta T_b, \quad (76)$$

where S_b denotes the SOC value when the trajectory reaches the temperature level T_b . Since T_b is a maximum temperature, this relation follows from $\dot{T} = 0$. For the 120°C boundary, where the simulation is stopped once the boundary is reached, this relation is used as a structural approximation.

From Eq. (76), the corresponding thermal boundary can be written as

$$U_{eq}(R_{isc}; T_b) = \frac{I_{ext}^2 R_{cell} + Q_{chem}(T_b)}{\Delta T_b} + \frac{\zeta U_{OCV}^2(S_b)}{\Delta T_b} \frac{1}{R_{cell} + R_{isc}}. \quad (77)$$

Therefore, the thermal boundary also has the form

$$U_{eq} = A_b + B_b \frac{1}{R_{cell} + R_{isc}}, \quad (78)$$

where

$$A_b = \frac{I_{ext}^2 R_{cell} + Q_{chem}(T_b)}{\Delta T_b}, \quad B_b = \frac{\zeta U_{OCV}^2(S_b)}{\Delta T_b}. \quad (79)$$

Equations (73) and (78) show the main structural reason for the similarity between the two types of curves. Both contain a constant term and an inverse resistance term. The inverse resistance term comes from the internal short circuit contribution.

The conclusion is that the time scale parameter and the thermal boundary share the same main resistance dependence. This explains why the simulated thermal boundary may be close to a constant ϵ curve in the (R_{isc}, U_{eq}) plane. The actual representative value of ϵ must be obtained from the full two dimensional ODE simulation.

3.3.2 Simulation based thermal boundaries and representative time scale values

The structural argument above suggests that a simulated thermal boundary may be organized by the time scale parameter. This subsection checks this directly using the full two dimensional ODE model. All numerical simulations in this thesis were performed in Python. For each fixed parameter pair (R_{isc}, U_{eq}) , the full two-dimensional T-SOC ODE system was integrated from the initial condition $S_0 = 1$ and $T_0 = T_{amb}$. The simulation was stopped when the temperature reached 120°C , when the SOC was depleted, or when the final simulation time was reached. The maximum temperature $T_{\max}(R_{isc}, U_{eq})$ was recorded for each parameter pair. The thermal boundaries $T_{\max} = 70^\circ\text{C}$ and $T_{\max} = 120^\circ\text{C}$ were then extracted from the resulting temperature map.

For each parameter pair (R_{isc}, U_{eq}) , the full model is simulated from the prescribed initial condition. The maximum temperature reached during the early stage simulation is denoted by

$$T_{\max}(R_{isc}, U_{eq}) = \max_{0 \leq t \leq t_{end}} T(t). \quad (80)$$

The simulation is stopped once the temperature reaches 120°C , because the present model only interprets before the irreversible chain reaction stage. Thus, $T_{\max}$ in this subsection should be understood as the maximum temperature within the validity range of the early stage model.

For a prescribed temperature level T_b , the simulated thermal boundary is defined as

$$\Gamma_{T_b} = \{(R_{isc}, U_{eq}) : T_{\max}(R_{isc}, U_{eq}) = T_b\}. \quad (81)$$

In this thesis, two temperature levels are considered. The first one is

$$T_b = 343.15 \text{ K}, \quad (82)$$

corresponding to 70°C , which is used as a BMS warning threshold. The second one is

$$T_b = 393.15 \text{ K}, \quad (83)$$

corresponding to 120°C , which is used as the boundary beyond which irreversible chain reactions are assumed to become significant.

To verify the relationship between a simulated thermal boundary and the time scale parameter, ϵ is evaluated along the boundary. Let

$$(R_{isc,i}, U_{eq,i}) \in \Gamma_{T_b}, \quad i = 1, \dots, N_b, \quad (84)$$

be the sampled points on a simulated boundary. The corresponding time scale values are

$$\epsilon_i = \epsilon(R_{isc,i}, U_{eq,i}). \quad (85)$$

The representative time scale value of this boundary is defined as

$$\bar{\epsilon}_{T_b} = \frac{1}{N_b} \sum_{i=1}^{N_b} \epsilon_i. \quad (86)$$

The relative variation of ϵ along the boundary is measured by

$$E_{\epsilon, T_b} = \frac{\sqrt{\frac{1}{N_b} \sum_{i=1}^{N_b} (\epsilon_i - \bar{\epsilon}_{T_b})^2}}{\bar{\epsilon}_{T_b}}. \quad (87)$$

A smaller value of E_{ϵ, T_b} means that the simulated thermal boundary is closer to a single constant ϵ curve.

The above analysis is then applied to the simulated 70°C boundary and the 120°C boundary. For the 70°C boundary, $N_b = 117$ sampled points are obtained. The mean and median values of the time scale ratio are

$$\bar{\epsilon}_{70} = 0.3938, \quad \text{median}(\epsilon_i) = 0.3310. \quad (88)$$

The sampled values satisfy

$$0.3199 \leq \epsilon_i \leq 1.4421, \quad (89)$$

and the relative variation is

$$E_{\epsilon, 70} = 41.56\%. \quad (90)$$

The sampled boundary covers the range

$$0.01 \leq R_{isc} \leq 1.3091 \Omega, \quad 0.05 \leq U_{eq} \leq 9.7336 \text{ W/K}. \quad (91)$$

Although the relative variation is still not small, the simulated 70°C boundary remains close to the corresponding mean ϵ curve in the parameter plane. Therefore, in an averaged sense, the 70°C thermal boundary can be represented by the characteristic value

$$\epsilon \approx 0.394. \quad (92)$$

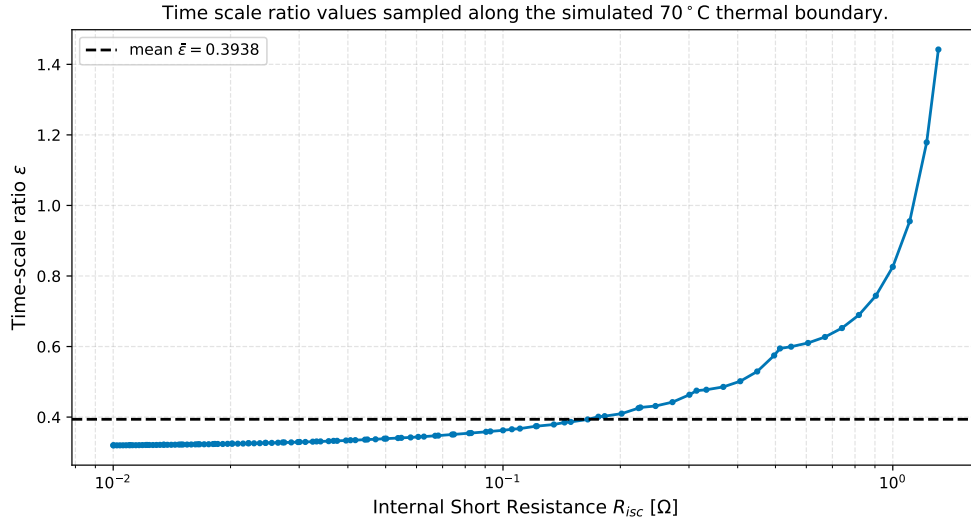


Figure 9: Time scale ratio values sampled along the simulated 70°C thermal boundary. The dashed line shows the mean value $\bar{\epsilon}_{70} = 0.3938$.

Figure 9 shows the sampled values of ϵ_i along the 70°C boundary. For most of the sampled boundary, the values stay close to the main range around 0.32 to 0.35. Only the endpoint region shows a visible upward deviation. The interpretation of these endpoint deviations is discussed together with the 120°C boundary below.

For the 120°C thermal boundary, $N_b = 46$ sampled points are obtained. The mean and median values are

$$\bar{\epsilon}_{120} = 1.8371, \quad \text{median}(\epsilon_i) = 1.6720. \quad (93)$$

The sampled values satisfy

$$1.5217 \leq \epsilon_i \leq 3.9961, \quad (94)$$

and the relative variation is

$$E_{\epsilon, 120} = 25.16\%. \quad (95)$$

The sampled boundary covers the range

$$0.01 \leq R_{isc} \leq 0.2345 \Omega, \quad 0.05 \leq U_{eq} \leq 1.9967 \text{ W/K}. \quad (96)$$

Thus, the value $\epsilon \approx 1.84$ gives the mean time scale level corresponding to the simulated 120°C thermal boundary.

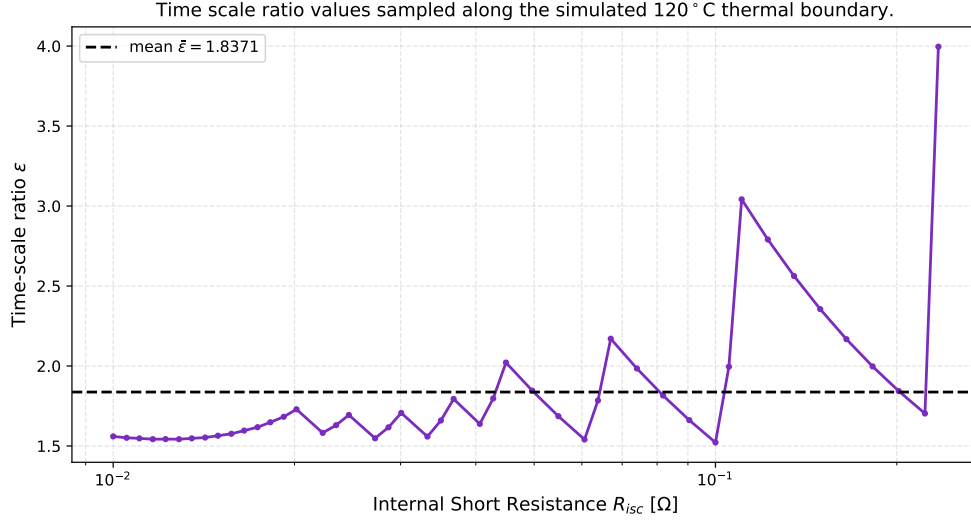


Figure 10: Time scale ratio values sampled along the 120°C thermal boundary. The dashed line shows the mean value $\bar{\epsilon}_{120} = 1.8371$.

Figure 10 shows that the sampled values along the 120°C boundary are less uniform than those along the 70°C boundary. The main deviation appears near the right end of the sampled curve. This deviation is not caused by a sudden physical change in the time scale structure, but mainly by the boundary extraction and sampling procedure.

Near this endpoint, the extracted 120°C boundary approaches the lower edge of the scanned U_{eq} range, and the last sampled point reaches $U_{eq} = 0.05 \text{ W/K}$. Therefore, the corresponding ϵ_i value is more sensitive to the finite computational domain and the discrete contour extraction. A similar but weaker endpoint effect also appears along the 70°C boundary. Thus, the endpoint shifts in the sampled ϵ_i curves should be interpreted as numerical sampling effects, not as a new physical mechanism.

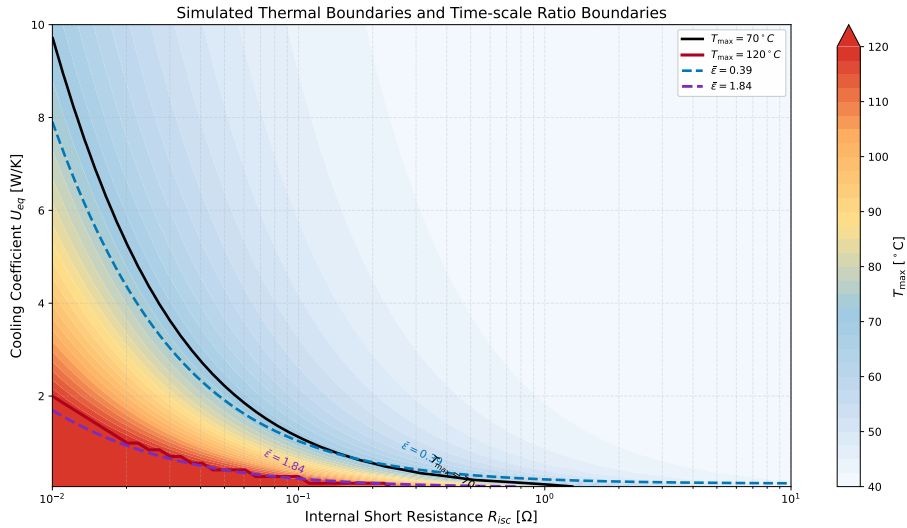


Figure 11: Simulated thermal boundaries and corresponding mean time scale ratio values. The black and red solid curves show the simulated 70°C boundary and the 120°C boundary, respectively. The dashed curves show the constant ϵ values using the mean values evaluated along the corresponding boundaries.

Figure 11 compares the simulated thermal boundaries with the corresponding mean ϵ values in the (R_{isc}, U_{eq}) parameter plane. The 70°C boundary is represented by the mean value $\epsilon = 0.39$, while the 120°C boundary follows the overall trend of $\epsilon = 1.84$. The agreement is not an exact quantitative equivalence, especially near the low U_{eq} end of the scanned domain. Still, the result shows a clear geometric correspondence between the simulated thermal boundaries and the time scale ratio.

Overall, the comparison shows that the time scale ratio ϵ is strongly related to the simulated thermal boundaries, but this relation should be interpreted together with the extracted boundary location. For the 70°C boundary, the sampled ϵ_i values increase gradually with R_{isc} , and the main deviation appears only near the final low cooling part of the boundary. For the 120°C boundary, the last large deviation is mainly caused by the fact that the stopping boundary reaches the lower limit of the scanned U_{eq} range. Therefore, ϵ should be used as an averaged geometric indicator, not as a pointwise exact replacement for the simulated thermal boundary. Once an internal short resistance R_{isc} is diagnosed and the current cooling condition U_{eq} is estimated, the corresponding point in the (R_{isc}, U_{eq}) parameter plane can still be evaluated through the time scale ratio. In this way, the model gives an early indication of whether the system is likely to remain near the warning level boundary or move toward the high temperature stopping boundary.

4 Numerical Sensitivity Analysis

The previous chapter analyzed the relation between the simulated thermal boundaries and the time scale ratio under a fixed operating condition. However, in a real battery system, the ambient temperature and the external current are not fixed. Both factors can change the heat balance and may shift the position of the thermal boundaries in the (R_{isc}, U_{eq}) parameter plane. Therefore, this chapter studies whether the relation between the thermal boundaries and the time scale ratio remains useful when these external conditions are changed.

Two sensitivity studies are considered. First, the effect of ambient temperature is analyzed by comparing $T_{amb} = 20^\circ\text{C}$, 40°C , and 60°C . Second, the effect of external current is studied by comparing $I_{ext} = 0\text{A}, 2.5\text{A}, 5\text{A}, 7.5\text{A}$ and 25A . The case $T_{amb} = 40^\circ\text{C}$ and $I_{ext} = 5\text{A}$ is used as the reference case, since it was already discussed in detail in the previous chapter.

The purpose of this chapter is not to repeat the full analysis for every parameter setting. Instead, the goal is to check how robust the boundary interpretation is when the operating condition changes. In particular, the following questions are considered: how the 70°C warning boundary and the 120°C high temperature boundary move, how the corresponding mean time scale ratio changes, and whether the time scale ratio still gives a useful geometric description of the thermal tendency.

4.1 Effect of Ambient Temperature

In this subsection, the effect of ambient temperature is examined by changing only T_{amb} , while all the other model parameters are kept the same. The three ambient temperatures considered are

$$T_{amb} = 20^\circ\text{C}, \quad 40^\circ\text{C}, \quad 60^\circ\text{C}. \quad (97)$$

For each value of T_{amb} , the (R_{isc}, U_{eq}) parameter plane is scanned. From these simulations, the two thermal boundaries corresponding to $T_{\max} = 70^\circ\text{C}$ and $T_{\max} = 120^\circ\text{C}$ are then extracted.

In the thermal equation, the ambient temperature appears in the heat dissipation term

$$U_{eq}(T - T_{amb}). \quad (98)$$

Therefore, when the cell temperature T is fixed, increasing T_{amb} reduces the temperature difference $T - T_{amb}$.

This means that the effective heat dissipation becomes weaker when the surrounding environment is hotter. So, intuitively, a higher ambient temperature should make the system easier to approach the warning and high temperature regions.

Figure 12 shows the simulated thermal boundaries under the three ambient temperatures. The result follows this expected trend. When T_{amb} increases from 20°C to 60°C , both the 70°C and 120°C boundaries move toward larger cooling coefficients. In other words, for the same internal short resistance R_{isc} , stronger cooling is required to keep the maximum temperature below the same threshold.

The corresponding mean time scale ratio curves are shown in Figure 13. The mean values sampled along the thermal boundaries are

$$\bar{\epsilon}_{70} = 0.767, \quad \bar{\epsilon}_{120} = 5.070, \quad T_{amb} = 20^\circ\text{C}, \quad (99)$$

$$\bar{\epsilon}_{70} = 0.394, \quad \bar{\epsilon}_{120} = 1.837, \quad T_{amb} = 40^\circ\text{C}, \quad (100)$$

and

$$\bar{\epsilon}_{70} = 0.159, \quad \bar{\epsilon}_{120} = 0.960, \quad T_{amb} = 60^\circ\text{C}. \quad (101)$$

These values show that the representative time scale ratio is not fixed. It changes with the ambient temperature because the thermal boundary itself moves to a different part of the (R_{isc}, U_{eq}) plane.

One point needs to be interpreted carefully. A higher ambient temperature gives smaller mean ϵ values in this set of simulations, but this does not mean that the system becomes safer. The reason is that the higher ambient temperature shifts the thermal boundary toward larger U_{eq} . Since the time scale ratio contains the factor $1/U_{eq}$, sampling ϵ along this shifted boundary can give a smaller mean value. Therefore, $\bar{\epsilon}$ should be understood as a representative value of the extracted boundary, not as a direct safety ranking by itself.

Because Figure 13 contains all ambient temperature cases at the same time, the comparison is a bit crowded. Figures 14 and 15 show the 20°C and 60°C cases separately, using the same format as the reference case.

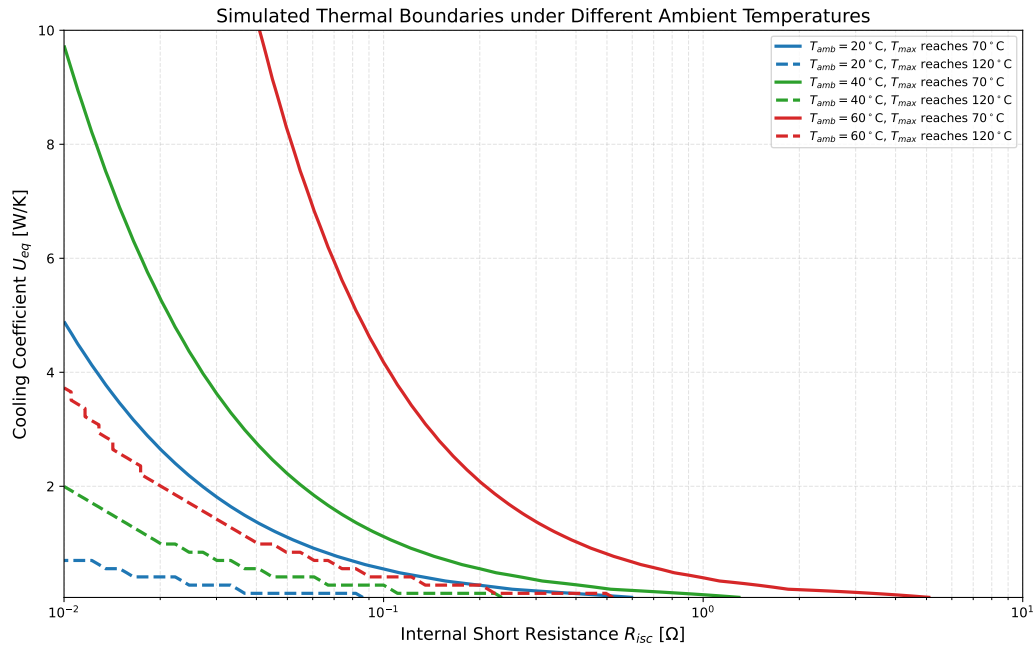


Figure 12: Simulated thermal boundaries under different ambient temperatures. The solid curves represent the 70°C boundary, and the dashed curves represent the 120°C boundary.

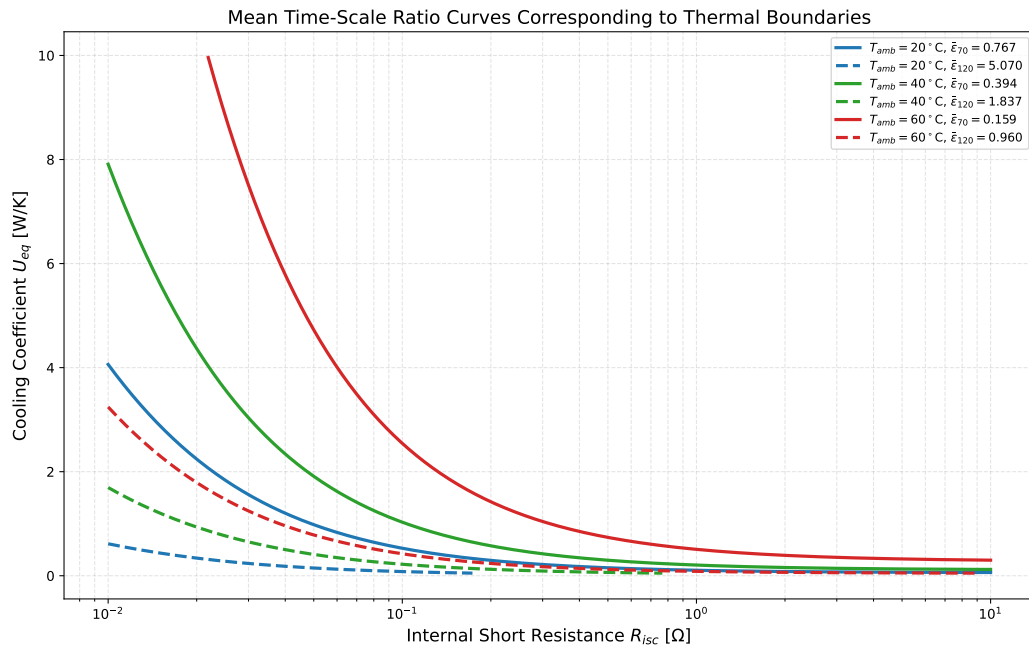


Figure 13: Mean time scale ratio curves corresponding to the simulated thermal boundaries under different ambient temperatures.

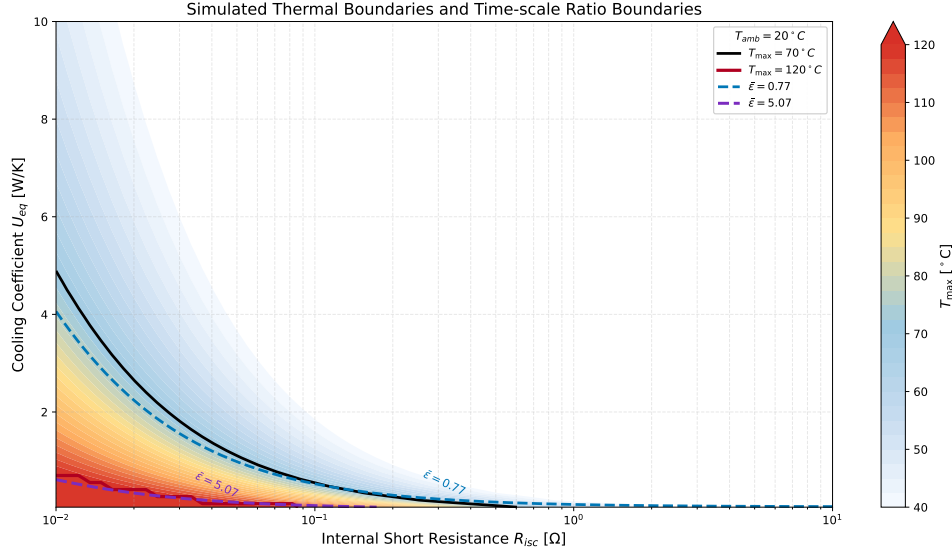


Figure 14: Simulated thermal boundaries and corresponding mean time scale ratio curves for $T_{amb} = 20^{\circ}\text{C}$.

This makes the relation between the simulated thermal boundaries and their corresponding mean ϵ curves clearer.

For $T_{amb} = 20^{\circ}\text{C}$, the 70°C boundary is represented by the mean value $\bar{\epsilon}_{70} = 0.77$, while the 120°C boundary is represented by $\bar{\epsilon}_{120} = 5.07$. The two mean ϵ curves still follow the overall locations of the simulated thermal boundaries. The colder environment allows the system to tolerate weaker cooling before reaching the same temperature thresholds, so the boundaries appear lower in the parameter plane.

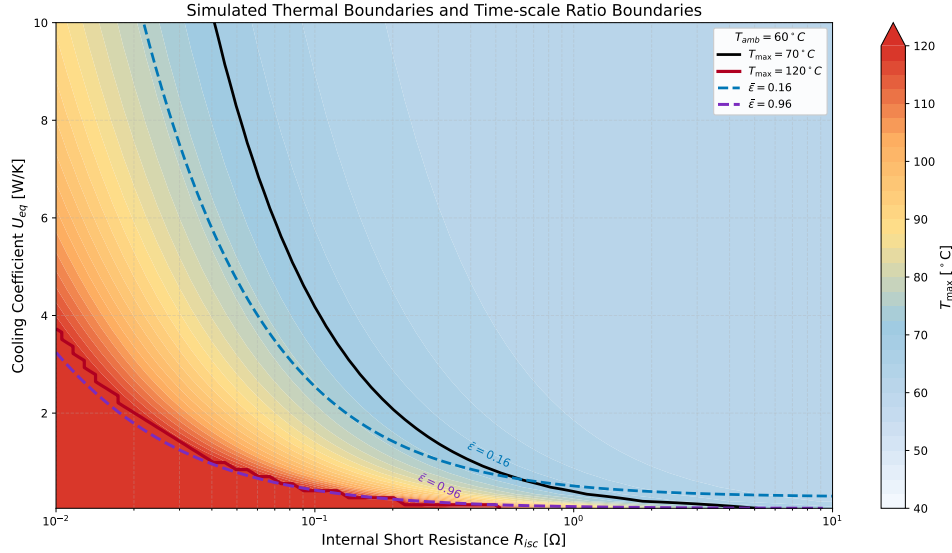


Figure 15: Simulated thermal boundaries and corresponding mean time scale ratio curves for $T_{amb} = 60^{\circ}\text{C}$.

For $T_{amb} = 60^{\circ}\text{C}$, the thermal boundaries move upward. The mean values become $\bar{\epsilon}_{70} = 0.16$ and $\bar{\epsilon}_{120} = 0.96$. The smaller representative values are mainly a consequence of the larger U_{eq} values along the extracted boundaries. The physical meaning is still consistent: the hotter environment reduces the effective heat dissipation, so stronger cooling is needed to avoid reaching the same temperature thresholds.

Overall, the ambient temperature mainly changes the thermal balance through the heat dissipation term. A colder environment shifts the thermal boundaries toward weaker cooling, while a hotter environment shifts them toward stronger cooling. The time scale ratio curves still provide a useful geometric description of the thermal boundaries, but their representative mean values depend on the ambient temperature.

4.2 Sensitivity to External Current

In this subsection, the ambient temperature is fixed at

$$T_{amb} = 40^\circ\text{C}, \quad (102)$$

and all other model parameters are kept unchanged. Only the external current is varied, with

$$I_{ext} = 0, 2.5, 5, 7.5, 25 \text{ A}. \quad (103)$$

It should be noted that the 25 A case corresponds to 1C and is used here only as a larger discharge current sensitivity case to test how far the time scale based interpretation remains consistent with the simulated thermal boundaries. Very high discharge currents or fast charging currents beyond this range are not fully represented by the present constant current discharge model and should be investigated in future work with more realistic current profiles.

Figure 16 compares the simulated thermal boundaries under different external currents. It can be seen that changing I_{ext} does not significantly move the thermal boundaries. The 70°C boundary is almost overlapped for all current cases, and the 120°C boundary only shows slight fluctuations. This is different from the ambient temperature sensitivity in the previous subsection, where the boundary shift is much more visible. Thus, in the current range considered here, the external current has only a limited influence on T_{max} , and it is not the dominant factor controlling the thermal boundary location.

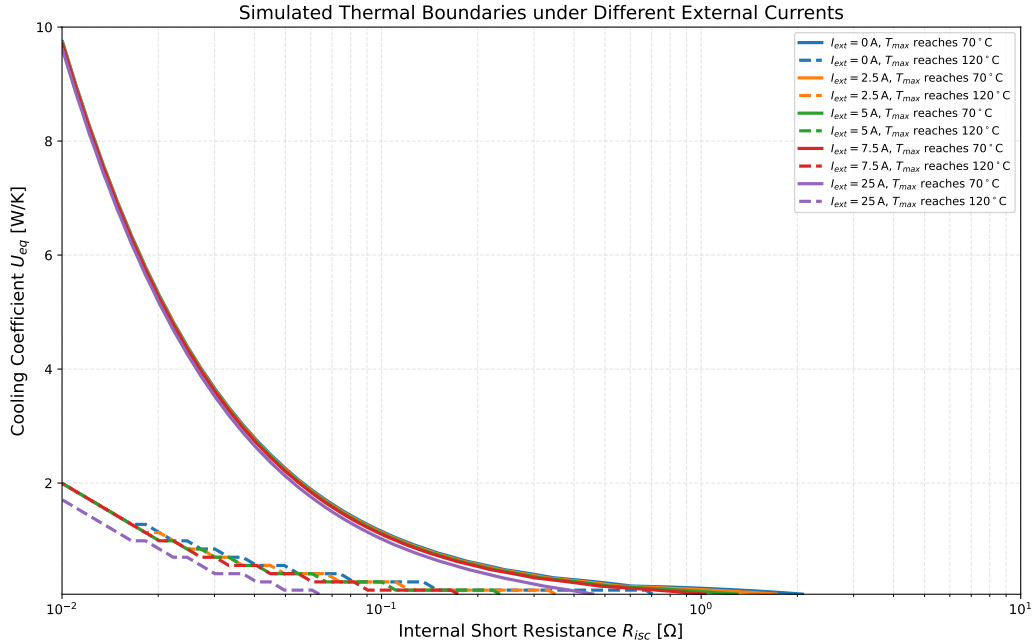


Figure 16: Simulated thermal boundaries under different external currents at $T_{amb} = 40^\circ\text{C}$. The solid curves represent the $T_{max} = 70^\circ\text{C}$ boundary, and the dashed curves represent the 120°C thermal boundary.

This weaker influence can be understood from the model itself. The external current contributes to the heat generation through

$$Q_{ext} = I_{ext}^2 R_{cell}. \quad (104)$$

Since $R_{cell} = 0.002 \Omega$, for $I_{ext} = 7.5 \text{ A}$, the external current heat is only

$$Q_{ext} = 7.5^2 \times 0.002 \approx 0.113 \text{ W}. \quad (105)$$

Even for the larger discharge current case $I_{ext} = 25 \text{ A}$, the corresponding value is

$$Q_{ext} = 25^2 \times 0.002 = 1.25 \text{ W}. \quad (106)$$

This value is still small compared with the internal short circuit heat in the low R_{isc} region. For example, using $U_{ref} \approx 4.19 \text{ V}$ and $\zeta = 0.28$, the reference internal short circuit heat can be estimated as

$$Q_{isc}^{ref} = \zeta \frac{U_{ref}^2}{R_{cell} + R_{isc}}. \quad (107)$$

When $R_{isc} = 0.1 \Omega$,

$$Q_{isc}^{ref} = 0.28 \frac{4.19^2}{0.002 + 0.1} \approx 48.2 \text{ W}, \quad (108)$$

and when $R_{isc} = 0.01 \Omega$,

$$Q_{isc}^{ref} = 0.28 \frac{4.19^2}{0.002 + 0.01} \approx 409.6 \text{ W}. \quad (109)$$

Therefore, even in the 25 A case, the direct heat generated by the external current is much smaller than the internal short circuit heat in the high risk low R_{isc} region. This explains why increasing I_{ext} only slightly changes the thermal boundary and does not strongly enlarge the high temperature region.

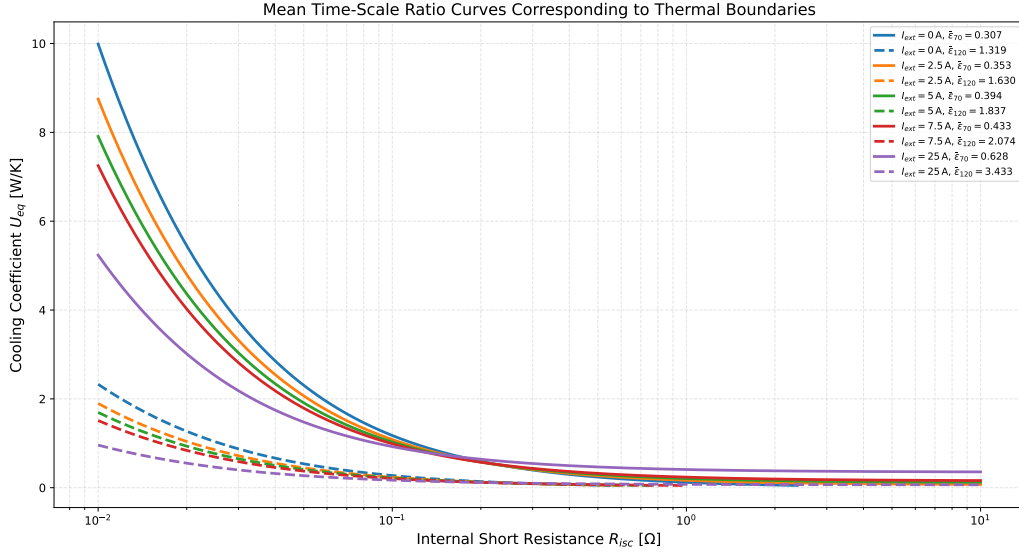


Figure 17: Mean time scale ratio curves corresponding to the thermal boundaries under different external currents at $T_{amb} = 40^\circ\text{C}$. The reference curves $\epsilon = 0.1$ and $\epsilon = 0.3$ are plotted using the $I_{ext} = 5 \text{ A}$ case.

Figure 17 shows the mean time scale ratio curves corresponding to the thermal boundaries. The mean value along the 70°C boundary increases from

$$\bar{\epsilon}_{70} = 0.307 \quad \text{at} \quad I_{ext} = 0 \text{ A} \quad (110)$$

to

$$\bar{\epsilon}_{70} = 0.433 \quad \text{at} \quad I_{ext} = 7.5 \text{ A}. \quad (111)$$

For the larger current case $I_{ext} = 25 \text{ A}$, it further increases to

$$\bar{\epsilon}_{70} = 0.628 \quad \text{at} \quad I_{ext} = 25 \text{ A}. \quad (112)$$

The mean value along the 120°C boundary also increases from

$$\bar{\epsilon}_{120} = 1.319 \quad \text{at} \quad I_{ext} = 0 \text{ A} \quad (113)$$

to

$$\bar{\epsilon}_{120} = 2.074 \quad \text{at} \quad I_{ext} = 7.5 \text{ A}, \quad (114)$$

and reaches

$$\bar{\epsilon}_{120} = 3.433 \quad \text{at} \quad I_{ext} = 25 \text{ A}. \quad (115)$$

Figures 18, 19, 20, and 21 show the individual thermal maps for $I_{ext} = 0, 2.5, 7.5$, and 25 A , respectively. The 5 A case is not repeated here, since it has already been given in Figure 11. These maps confirm the same trend. As the external current increases, the corresponding mean ϵ values become larger, while the high temperature region does not expand significantly. In the 25 A case, the time scale ratio curves become less accurate as a geometric approximation of the simulated thermal boundaries. However, the values corresponding to the 70°C and 120°C boundaries are still clearly separated. This means that the time scale ratio still gives a useful distinction between the warning boundary and the higher temperature boundary, although its boundary fitting accuracy decreases at large external current.

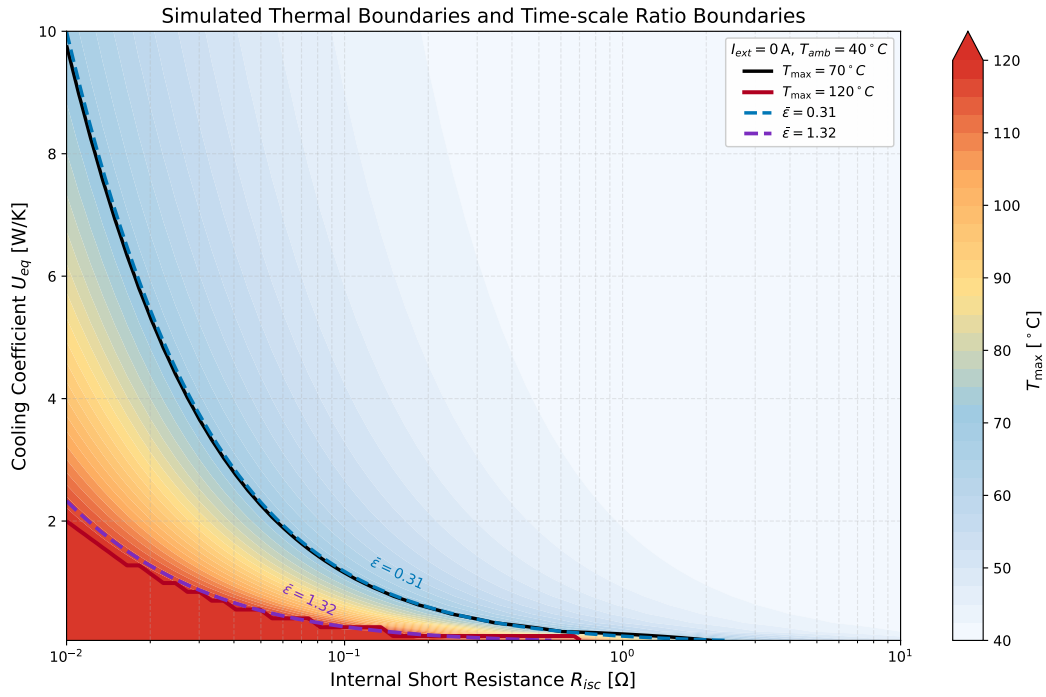


Figure 18: Simulated thermal boundaries and corresponding mean time scale ratio curves for $I_{ext} = 0 \text{ A}$ at $T_{amb} = 40^\circ \text{C}$.

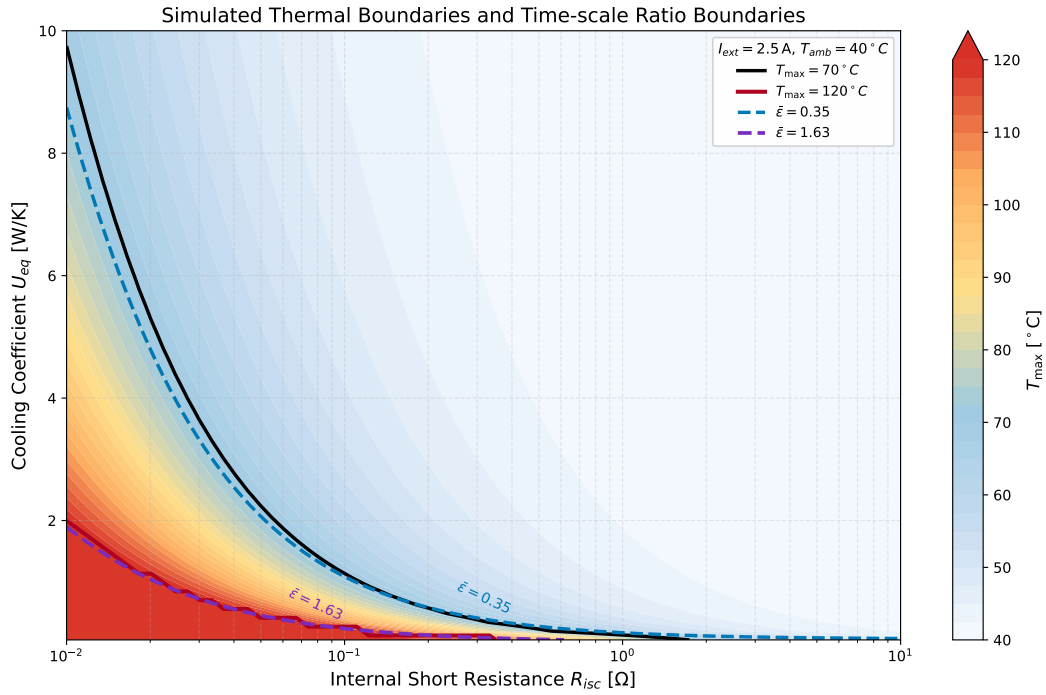


Figure 19: Simulated thermal boundaries and corresponding mean time scale ratio curves for $I_{ext} = 2.5 \text{ A}$ at $T_{amb} = 40^\circ \text{C}$.

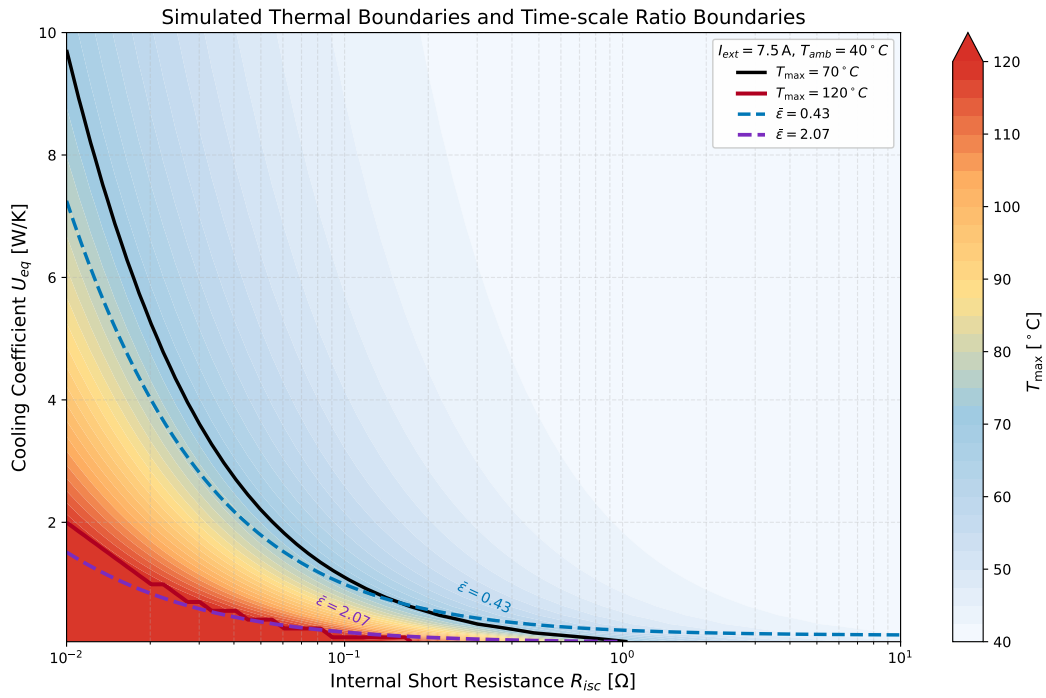


Figure 20: Simulated thermal boundaries and corresponding mean time scale ratio curves for $I_{ext} = 7.5 \text{ A}$ at $T_{amb} = 40^\circ\text{C}$.

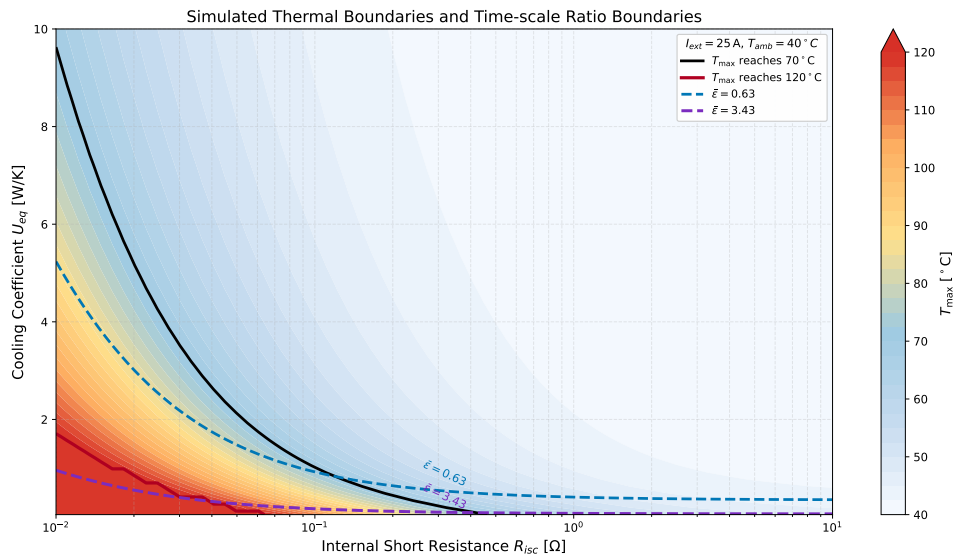


Figure 21: Simulated thermal boundaries and corresponding mean time scale ratio curves for $I_{ext} = 25 \text{ A}$ at $T_{amb} = 40^\circ\text{C}$.

This behavior can be understood from the definition of the time scale ratio. According to the definition used in this thesis,

$$\epsilon(R_{isc}, U_{eq}) = \frac{mC_p}{3600Q_{cap}U_{eq}} \left(I_{ext} + \frac{U_{ref}}{R_{cell} + R_{isc}} \right). \quad (116)$$

For a fixed value $\epsilon = \epsilon_c$, the corresponding curve in the (R_{isc}, U_{eq}) plane is

$$U_{eq}^{\epsilon_c} = \frac{mC_p}{3600Q_{cap}\epsilon_c} \left(I_{ext} + \frac{U_{ref}}{R_{cell} + R_{isc}} \right). \quad (117)$$

When $I_{ext} = 0$, this curve is mainly controlled by the inverse resistance term

$$\frac{1}{R_{cell} + R_{isc}},$$

which has a parameter dependence similar to that of the internal short circuit heat generation. As shown in Figure 18, the fitted time scale curve can follow the simulated thermal boundary relatively well. However, as I_{ext} increases, the constant current term in Eq. (117) becomes more important. The relative contribution of the inverse resistance term is then weakened, and the fitted ϵ curve is no longer governed mainly by R_{isc} . This is why the agreement between the mean time scale ratio curve and the simulated thermal boundary becomes worse when I_{ext} is increased, especially in the 25 A case shown in Figure 21.

This point is important for interpreting the current result. The thermal boundary itself is almost unchanged because the direct heat contribution $I_{ext}^2 R_{cell}$ is still small compared with the internal short circuit heat. In contrast, the time scale ratio is more sensitive to I_{ext} , because the external current directly accelerates SOC depletion. Therefore, at larger external current, ϵ still reflects the relative speed between temperature and SOC dynamics, but it becomes less accurate as a geometric approximation of the thermal boundary.

Overall, the external current sensitivity test shows that increasing I_{ext} mainly increases the time scale ratio and accelerates SOC depletion, but it does not significantly enlarge the high temperature region. The 25 A case makes this point clearer. Although the time scale ratio curves become less accurate in matching the simulated thermal boundaries, the 70°C and 120°C boundaries still correspond to clearly different time scale levels. Therefore, in the present parameter range, the thermal runaway boundary is mainly controlled by internal short resistance, cooling strength, and ambient temperature. The external current mainly modifies the SOC depletion rate and the effective time scale separation, rather than directly controlling T_{max} .

5 Conclusions

5.1 Main Findings

This thesis developed an interpretable early stage thermal safety assessment framework for internal short circuit induced thermal runaway in lithium ion batteries. The main purpose is to understand how the early thermal response is affected by internal short circuit resistance, heat dissipation capability, ambient temperature, external current, and the coupled evolution of state of charge. Different from a fixed temperature threshold method, the proposed framework treats the temperature and SOC evolution as a two dimensional nonlinear dynamical system, and uses the parameter location in the (R_{isc}, U_{eq}) plane to interpret the possible thermal tendency of the cell.

The first contribution of this thesis is the construction of a two dimensional temperature and SOC model for early stage internal short circuit analysis. The model includes internal short circuit ohmic heat, external current heat, early SEI and anode reaction heat, and equivalent linear heat dissipation. In this model, R_{isc} controls the internal short circuit current, SOC depletion rate, and internal short circuit heat generation, while U_{eq} controls the strength of heat dissipation to the environment. Therefore, these two parameters provide a natural parameter plane for discussing thermal safety boundaries.

Based on this model, a parameter dependent time scale ratio $\epsilon(R_{isc}, U_{eq})$ was derived from the nominal thermal response time and the SOC depletion time. This ratio is useful because it does not only describe whether a fast slow structure exists, but also helps decide which type of mathematical interpretation is reasonable in different parameter regions. When ϵ is small, the temperature response is much faster than SOC depletion and the system is closer to a fast slow system. When ϵ becomes larger, the temperature and SOC evolve on closer time scales, so a strict fast slow reduction is no longer appropriate.

The analysis of the strong time scale separation region shows that the fast subsystem, the critical manifold, and the fast thermal attraction condition can be formulated clearly in the present model. For the parameter range considered in this thesis, the conservative temperature check indicates that the strongly separated region remains below the warning threshold under the present assumptions. Therefore, the role of the strong separation analysis is mainly to clarify the fast thermal balance structure and to identify when the temperature direction is locally attracting.

For the weak and non fast slow regions, the thesis used a Jacobian based local thermal direction stability criterion. The local thermal growth rate $\lambda_2(T)$ was used to determine whether a small temperature perturbation is locally damped or amplified along a trajectory. This gives an additional state based interpretation of the thermal response. The temperature thresholds describe how high the cell temperature has reached, while $\lambda_2(T)$ describes whether the local thermal direction is still damping or has become amplifying.

The main result of this thesis is the comparison between simulated thermal boundaries and constant ϵ curves. The simulated 70°C thermal boundary and 120°C thermal boundary show a clear geometric relation with the time scale ratio in the (R_{isc}, U_{eq}) plane. This relation can be explained by the model structure, because both the time scale ratio and the approximate thermal balance boundary contain a similar dependence on the inverse resistance term $1/(R_{cell} + R_{isc})$. Therefore, the time scale ratio provides a useful way to interpret why the thermal boundary has its observed shape in the parameter plane.

The sensitivity analysis further showed that ambient temperature has a strong influence on the thermal boundary location. When T_{amb} increases, the temperature difference $T - T_{amb}$ becomes smaller at the same cell temperature. This weakens the effective heat dissipation, so stronger cooling is required to keep the cell below the same warning or stopping threshold. As a result, both the 70°C and 120°C boundaries move toward larger U_{eq} values when the ambient temperature increases.

The influence of external current is different. In the current range considered in this thesis, increasing I_{ext} does not significantly move the simulated thermal boundaries. This is because the direct heat contribution $I_{ext}^2 R_{cell}$ is small compared with the internal short circuit heat in the low R_{isc} region. However, the external current still affects the SOC depletion rate and therefore changes the time scale ratio. This means that external current mainly modifies the effective time scale separation, while the maximum temperature boundary is still mainly controlled by internal short circuit resistance, cooling strength, and ambient temperature.

Overall, the thesis shows that early stage thermal risk under internal short circuit conditions should not be judged only by a fixed temperature threshold. The same temperature level may correspond to different later tendencies depending on R_{isc} , U_{eq} , SOC, ambient temperature, and external current. The proposed framework connects these factors through a two dimensional dynamical model, a time scale ratio, local thermal direction stability, and simulated thermal boundaries. Once the internal short circuit resistance and cooling condition

are estimated by upstream BMS algorithms, the corresponding location in the parameter plane can provide an interpretable indication of whether the cell is likely to remain below the warning boundary or move toward a high temperature region.

5.2 Limitations and Future Work

The present work also has several limitations. First, the battery is represented by a lumped parameter thermal model. This assumption makes the model suitable for fast parameter scanning, but it cannot describe local hot spots, spatial temperature gradients, or the detailed position of the internal short circuit inside the cell. In real batteries, these effects may be important, especially when the short circuit is highly localized. High speed imaging studies have shown that internal structural damage and local thermal behavior can strongly affect the initiation and propagation of thermal runaway^[15].

Second, the chemical heat generation model is limited to early stage SEI and anode side reactions. This is consistent with the scope of the thesis, because the model is intended for the early warning range before the cell enters severe thermal runaway. However, the model should not be used to predict the full high temperature chain reaction after 120°C. More detailed reaction kinetics would be needed if the later runaway process is to be described.

Third, R_{isc} and U_{eq} are treated as given parameters in this thesis. This is reasonable for studying how the system behaves once the internal short circuit severity and cooling condition have been estimated. However, the model does not describe how the internal short circuit forms, how R_{isc} evolves with time, or how the cooling condition changes under real operating conditions.

Fourth, the conclusions are based on numerical simulations and parameter values taken from literature based models. Experimental validation is still needed. In particular, future work should test whether the extracted 70°C and 120°C boundaries, the representative time scale ratio values, and the local thermal direction stability remain useful when compared with experimental internal short circuit data.

Future work can therefore proceed in several directions. A multi node or spatial thermal model could be introduced to include local hot spot effects while still keeping part of the dynamical systems interpretation. The internal short circuit resistance could be modeled as a time varying quantity to describe fault growth. More complete reaction kinetics could also be added for the transition from early stage warning to later thermal runaway. Finally, the proposed parameter plane criterion could be integrated with BMS state estimation and fault diagnosis algorithms, so that the model can be used not only for offline analysis, but also as part of an online early warning strategy.

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